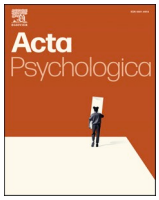
Acta Psychologica 259 (2025) 105374



Contents lists available at ScienceDirect

Acta Psychologica

journal homepage: www.elsevier.com/locate/actpsy





The impact of smartphone usage frequency on university students' academic performance: A meta-analysis of moderating factors

Chia-Yen Hsieh

Department of Early Childhood Education, National Pingtung University, No. 4-18, Minsheng Rd., Pingtung City, Pingtung County 900391, Taiwan

ARTICLE INFO

Keywords:
Smartphone usage frequency
Academic performance
Meta-analysis
Moderating factors
Quality education

ABSTRACT

This meta-analysis investigates the impact of smartphone usage on university students' academic performance, with a focus on identifying moderating factors. A total of 45 studies were analyzed, revealing a small but statistically significant negative effect of smartphone usage frequency on academic performance ($r = -0.12$). Moderation analyses were conducted on variables such as the source of data, region, usage purpose, and terminological differences in measuring smartphone use. Results show that smartphone addiction and problematic use yield more pronounced negative impacts on academic outcomes compared to general usage measures. Furthermore, multitasking during class demonstrated the highest negative effect among smartphone-related behaviors. The study emphasizes the potential benefits of using objective data collection methods, such as app-based tracking, while acknowledging that self-reported measures can still offer valuable insights, though they may be influenced by recall bias. These findings call for targeted educational interventions, promoting information literacy and self-regulation in smartphone use, in order to mitigate the detrimental effects on academic performance. Future research should explore longitudinal designs and standardized measurement frameworks to provide a more comprehensive understanding of the relationship between smartphone use and academic success.

1. Introduction

Although the COVID-19 pandemic has passed, it has brought about numerous impacts and changes in daily life, one of which is a shift in attitudes toward distance education and digital learning. The pandemic forced students and educators to use digital communication tools, such as online learning platforms, smartphones, tablets, and computers, for ICT-based learning. Moreover, the adoption of digital tools has promoted the development of new teaching strategies, incorporating multimedia materials and interactive exercises, effectively engaging students in remote learning environments (Rajeshkumar & Rajakumari, 2024). Among these tools, smartphones have become an indispensable part of daily life due to their multifunctionality for information gathering, communication, education, and entertainment (Al-Khlaiwi et al., 2020; Haug et al., 2015). These features have made smartphones increasingly popular, with new functions being added (Suresh et al., 2021; Yang et al., 2021). It is evident that the multifunctionality of smartphone technology has had a significant impact on our lives and social relationships (Yi et al., 2016). The younger generation uses smartphones for various activities, such as learning, entertainment, internet access, social networking, and communication (Aker et al., 2017). Smartphone usage has become an integral part of our daily lives, changing our daily activities and habits (Bragazzi et al., 2019). However, in recent years, some researchers have raised concerns about the excessive use of such devices among students. They argue that problems with smartphone usage arise when this interferes with daily activities (e.g., Busch & McCarthy, 2021; Gutiérrez-Puertas et al., 2019; Panova & Carbonell, 2018). The newly introduced term 'nomophobia', derived from the expression 'non-mobile-phone phobia' (Yildirim & Correia, 2015) refers to the anxiety of users experiencing a dependence or addiction to smartphones if deprived of access to their devices (Dixit et al., 2010).

Nomophobia, or the fear of being without a mobile phone, is a prevalent issue among higher education students, impacting their psychological well-being and daily activities. Furthermore, research indicates that female students tend to experience higher levels of nomophobia compared to their male counterparts (Khayer et al., 2024). This may be related to female students' more frequent checking of their phones and their reliance on smartphones as a tool for social connection (Khayer et al., 2024). Additionally, smartphone addiction mediates the

E-mail address: banduna@nptu.edu.tw.

https://doi.org/10.1016/j.actpsy.2025.105374
Received 15 November 2024; Received in revised form 22 April 2025; Accepted 30 July 2025
Available online 11 August 2025
0001-6918/© 2025 The Author. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

relationship between social media addiction (e.g., Facebook use) and nomophobia, highlighting the interconnectedness of digital behaviors (Aydin & Kuş, 2023). This suggests that modern individuals place greater importance on social interactions through social media, which leads to increased anxiety when they leave home without their phones or when their phones are not nearby, ultimately impacting their mental health (Khayer et al., 2024). When the variable of age is included, an interaction effect between age and gender is observed. For example, Bulut and Sengul (2024) suggested that younger women may be more susceptible to these effects. Additionally, the report shows that first-year university students reported higher levels of nomophobia, possibly due to the transition to a new academic environment and increased smartphone usage for social integration (Aydin & Kuş, 2023). In this work, the issue we are most concerned about is whether excessive smartphone use affects university academic performance. One study found a significant statistical association between levels of nomophobia and academic performance, indicating that higher levels of nomophobia may be linked to lower academic performance (Nguyen et al., 2024). Other research suggests that the anxiety associated with nomophobia can interfere with students' focus and time management, potentially impacting their academic outcomes (Khayer et al., 2024).

Research findings indicate that using social networking apps for just a few minutes during study or sleep can have negative effects (Sohn, 2014). Problematic smartphone use has also been associated with cognitive impairment (Wilmer et al., 2017), poor sleep quality (Demirci et al., 2015), and depression (Geng et al., 2021). Limiting smartphone use has been shown to reduce depression and improve sleep quality (Olson et al., 2023). All of these factors may affect students' academic performance. However, researchers continue to debate the relationship between smartphone use and academic performance, many believing that the effective use of smartphones for learning can enhance students' academic performance. For example, Anshari et al. (2017) pointed out that the advantages of smartphones in educational environments include the portability of rich content and the facilitation of knowledge sharing and dynamic learning. Cheng et al. (2016) reported that using smartphones in the classroom can effectively improve college students' English learning outcomes. Traxler (2009) also indicated that mobile learning through smartphones can enhance learning effectiveness. Other studies have shown that smartphones are effective in personalized learning (Steel, 2012), self-regulated learning (Sha et al., 2012), classroom engagement (Wang et al., 2009), and second language acquisition (Thornton & Houser, 2005). The results of these studies highlight the educational advantages of smartphone use, by bridging the digital gap, facilitating homework completion, accelerating peer collaboration, providing quick access to information, and increasing vocabulary.

While some studies have confirmed that smartphones can aid students' learning, from another perspective, they have also demonstrated that students are becoming increasingly reliant on smartphones to accomplish education-related tasks. Yet, over-reliance on smartphones may lead to problematic behaviors or even behavioral addiction (Panova & Carbonell, 2018; Satchell et al., 2021). Some studies have confirmed the negative effects of excessive smartphone use on university students. For instance, Hawi and Samaha (2016) clearly pointed out the adverse impact of smartphone addiction on academic performance, and Lee et al. (2015) found a negative correlation between smartphone addiction and university students' academic performance.

Consequently, more and more people are expressing a desire to reduce smartphone use to avoid developing harmful behaviors and lifestyle habits (Olson et al., 2023). In addition to experts advocating for reduced smartphone usage, many users have become aware of their own problematic smartphone use. For example, Lukoff et al. (2018) found that nearly half of smartphone users in developed countries believe they use their phones too much, and many would prefer to spend their time doing other things. However, only half of those who attempted to reduce their smartphone usage reported success.

Due to the powerful functions and portability of smartphones, global usage has surged, leading to an increasing dependence on them. In 2019, 96.8 % of people globally were using mobile devices (Sarker et al., 2019). This widespread usage indicates that some individuals prefer smartphones over laptops and personal computers because they offer many powerful features for entertainment, watching sports events, online gaming, online banking, mobile payments, sending emails, instant messaging, online meetings, online shopping, social media, calendar synchronization, location synchronization, and completing work or school assignments (Nayak, 2018). Additionally, it is worth mentioning that many university-age students tend to rely on various synchronous or asynchronous technologies, including smartphones, to maintain social relationships (Guo et al., 2011). Moreover, students prefer to communicate through instant messaging apps. Studies have demonstrated that smartphone use impacts the social relationships of university students (e.g., Kang & Jung, 2014; Park & Lee, 2012). Karikoski and Soikkeli (2013) asserted that smartphones facilitate interpersonal interactions either through verbal conversations or by transmitting textual and graphical information. Furthermore, their use also stimulates the creation or reinforcement of social networks. Park and Lee (2012) explored the relationship between smartphone usage motivation, social relationships, and mental health, finding a positive correlation between smartphone use and social intimacy.

From the above findings, it is clear that a significant amount of research has pointed out that university students are becoming increasingly dependent on smartphones. Bianchi and Phillips (2005) noted that most young people find it difficult to control the amount of time and frequency they spend on their phones. Given the heavy reliance of university students on smartphones, exploring the relationship between smartphone dependence and academic performance has become crucial.

However, despite the importance of this issue, one of the challenges in quantifying problematic smartphone use globally is the overwhelming number of available measures (Olson et al., 2022). There are at least 78 different scales (Harris et al., 2020), many of which are well-correlated with one another (Olson et al., 2022), for assessing similar constructs related to behavioral addiction (Sohn et al., 2019). Nonetheless, these measures vary in their criteria for defining problematic behavior or addiction risk (De-Sola Gutiérrez et al., 2016), making it difficult to directly compare scores across different studies in a meta-analysis.

The most commonly cited measures for problematic smartphone use are the Smartphone Addiction Scale (SAS; Kwon, Lee, et al., 2013) and its short version (SAS-SV; Kwon, Kim, et al., 2013). As of September 2024, these two measures were cited over 4500 times on Google Scholar. These measures were developed in collaboration with clinicians and cover various components related to behavioral addiction: withdrawal, tolerance, loss of control, intense desire to use, neglect of other activities, and continued use despite harm (Sohn et al., 2019). Both scales include items such as ‘Because of my smartphone use, I find it hard to concentrate in class, while doing homework, or at work.’ Participants rate their agreement on a six-point Likert scale, with higher scores indicating more problematic smartphone use. The full version of the scale captures more shared variance compared to several other related measures, while the short version predicts clinical judgments of smartphone addiction (Kwon, Kim, et al., 2013).

In summary, we observe several interesting phenomena. First, there is inconsistency in the research. Over the past few years, studies on the impact of university students' reliance on smartphones for academic tasks on their academic performance have produced mixed results. Second, there is inconsistency in assessing smartphone usage frequency. Many studies in recent years have attempted to understand the frequency of smartphone use among university students to determine whether they are excessively dependent on their devices. As a result, terms such as ‘addiction,’ ‘over-reliance,’ ‘overuse,’ ‘compulsive mobile phone usage,’ ‘heavy smartphone use,’ ‘problematic mobile phone usage,’ and ‘nomophobia’ have appeared in the literature, but these

terms do not clearly convey the extent or severity of usage frequency to readers. Third, the complexity of individual smartphone use makes it difficult to assess, as smartphones serve multiple functions (e.g., taking photos, browsing the internet, social networking, making calls, using instant messaging, posting status updates, listening to music, and watching videos). This variety of uses complicates the ability to focus on specific behaviors in research, potentially leading to inconsistent findings. Fourth, the variety of measurement tools presents a challenge. Given the difficulty of defining smartphone usage frequency and the complexity of its related behaviors, the available measurement tools are also highly diverse. As mentioned earlier, while the SAS scale is the most widely used, there are at least 78 different related scales, and some researchers have developed their own instruments. These scales vary between norm-referenced and criterion-referenced measures, further complicating meta-analyses.

In this context, we aim to conduct a global meta-analysis of these measures. Therefore, our research objectives are as follows:

Objective 1: To examine the impact of smartphone usage frequency on university students' academic performance.
Objective 2: To identify the factors in the literature that influence the relationship between smartphone usage frequency and university students' academic performance.
Objective 3: To determine how differences in research designs and methodologies might affect the relationship between smartphone usage frequency and university students' academic performance.

Based on the research objectives outlined above, the questions to be answered in this study are:

Objective 1:
1. What is the direction of effect size (positive or negative) in previous studies on the relationship between smartphone usage frequency and university students' academic performance?
2. What is the magnitude of effect size in previous studies on the relationship between smartphone usage frequency and university students' academic performance?
3. What are the results of heterogeneity analysis for models examining the relationship between smartphone usage frequency and university students' academic performance in previous studies?

Objective 2:
1. What factors (i.e., subgroup analysis) have been identified in past studies as influencing the relationship between smartphone usage frequency and university students' academic performance?
2. What factors have been identified as moderating variables in past studies that affect the relationship between smartphone usage frequency and university students' academic performance?

Objective 3: How does the relationship between smartphone usage frequency and university students' academic performance vary based on research design and methodological characteristics?

2. Literature review

2.1. Defining and measuring smartphone overuse in relation to academic performance

The increasing prevalence and usage of smartphones have led many studies to propose different terminologies and perspectives on excessive smartphone use, defined as using smartphones beyond normal levels. For example, the term "smartphone addiction" as introduced by Lee et al. (2017) frames excessive use as an addictive behavior with psychological implications, often aligned with dependency and withdrawal symptoms; On the other hand, Kim and Byrne (2011) coined "compulsive mobile phone usage," which focuses on an inability to control smartphone engagement despite adverse outcomes; Meanwhile, Chen et al. (2016) and Park et al. (2015) utilized the terms "excessive smartphone use" and "heavy smartphone use," respectively, emphasizing quantitative aspects, such as time spent on smartphones, without necessarily implying dependency; De-Sola Gutiérrez et al. (2016) used the term "cell-phone addiction"; Kuss et al. (2018) and Yildirim and Correia (2015) introduced "problematic mobile phone usage" and "nomobile-phone phobia," emphasizing behavioral disruption and avoidance of discomfort from being without a phone.; Zhang et al. (2020) used "mobile phone dependence." These terminologies, though overlapping, reveal how different conceptual lenses influence the framing of smartphone overuse, shaping the research focus and potentially skewing findings. This concern regarding conceptual inconsistency has also been emphasized in recent systematic reviews, which call for the unification of terminology and clearer operational definitions to improve comparability across studies (Sanchez-Fernandez and Borda-Mas, 2023; Zhang & Zeng, 2024). As a result, studies examining academic performance may yield varying results, depending on how 'excessive use' is defined. A unified conceptualization is necessary to ensure methodological consistency and facilitate cross-study comparisons, enabling a more comprehensive understanding of the true impact of excessive smartphone use on student outcomes.

The distinction between smartphone addiction and problematic smartphone use is a critical aspect when conducting a meta-analysis on the impact of smartphone overuse on academic performance. Smartphone addiction is defined as a psychological dependency, characterized by compulsive use and withdrawal symptoms, which mirrors traditional addiction behaviors (Billieux et al., 2015). This severe level of dependency can lead to significant impairments in attention, cognitive functioning, and time management, all of which are crucial for academic success (Kuss & Griffiths, 2017). Conversely, problematic smartphone use refers to broader behaviors that negatively affect daily life without necessarily reaching the compulsive level of addiction. This broader category includes behaviors such as prolonged usage that interferes with academic activities, but may not have the same psychological and emotional depth as addiction (Panova & Carbonell, 2018). Consistent with this view, recent literature has highlighted the need for psychometrically validated tools that can distinguish between usage frequency, problematic behaviors, and psychological dependence to improve the reliability of findings (Kuş, 2025; Sunday et al., 2021; Zhang & Zeng, 2024). The distinction between these terms is vital for a meta-analysis, as inconsistent definitions across studies can obscure the true effects of smartphone overuse on academic performance. By establishing clear definitions, a meta-analysis can more effectively aggregate findings and identify patterns across diverse studies. The lack of a standardized definition of smartphone overuse poses a significant challenge for meta-analyses. Without consistency in terms, the findings from different studies may not be comparable, leading to misleading conclusions. Researchers have suggested that a unified approach, incorporating both quantitative measures—such as hours of daily use—and qualitative assessments of behaviors like loss of control and compulsivity, would create a more reliable basis for meta-analysis (Panova & Carbonell, 2018). Therefore, in line with research objective 2, we will specifically include 'smartphone usage terminology' in the subgroup analysis, and we will provide a detailed explanation of the criteria for inclusion in the meta-analysis within the 'Eligibility criteria' section.

Methodologies for measuring excessive smartphone use further underscore the importance of consistent metrics in meta-analysis. Studies employ diverse methods, from self-reported questionnaires like the SAS and SAS-SV, which measure behavioral symptoms, to usage logs that provide objective data on time spent using smartphones (Van Deursen et al., 2015). Some studies use mixed methods, combining both qualitative surveys and quantitative app usage data, offering a comprehensive understanding of smartphone use (Van Deursen et al., 2015). For a meta-analysis, harmonizing these methodologies is crucial to ensure that the aggregated data are comparable across studies. Due to the diversity of research methodologies, our third research objective is to explore in depth how different research designs and methods

influence the relationship between smartphone usage frequency and academic performance among university students.

By critically assessing how different studies define and measure smartphone overuse, researchers can account for inconsistencies, select appropriate studies for inclusion, and apply statistical adjustments if necessary. This thorough groundwork ensures that the meta-analysis is built on reliable and comparable data, ultimately leading to more robust and meaningful conclusions about the relationship between smartphone overuse and academic performance.

2.2. A literature review on the impact of smartphone usage frequency on academic performance

The relationship between smartphone usage frequency and academic performance has been widely studied, with multiple arguments in the existing literature suggesting both positive and negative associations between smartphone use and academic outcomes. However, as emphasized in the aforementioned literature, there is still no unified definition of smartphone usage frequency, particularly in terms of excessive use, and the behavior surrounding smartphone use is complex. Moreover, other factors, such as data collection methods and the specific measures of academic performance and smartphone use, also play a role. In this section, we review the main theoretical mechanisms.

In recent years, due to the rapid increase in the prevalence and usage frequency of smartphones, scientists from various disciplines have shown growing interest in the potential impacts of smartphone usage. As a result, scholars have studied the relationship between smartphone use and smartphone addiction (or what is referred to as overuse). For example, Demirci et al. (2015) explored the relationship between excessive smartphone use and sleep; Samaha and Hawi (2016) investigated the correlation between excessive smartphone use and life satisfaction; Chen and Peng (2008) discussed the relationship between college students' smartphone use and their social relationships; and Ho et al. (2014) examined the link between excessive smartphone use and Attention Deficit Hyperactivity Disorder (ADHD).

In addition to the aforementioned studies, smartphone use is also expected to interfere with individuals' educational and professional lives, particularly in relation to academic performance among college students. A systematic review highlighted a predominantly negative association between frequent smartphone use and academic success among university students (Amez & Baert, 2020), although this result may be influenced by differences in research design and the measurement methods employed. Variations in methodology may affect research outcomes; for instance, the results derived from self-reported data analysis may differ from those obtained using objective tools such as smartphone usage tracking. Another study found that increased daily smartphone usage leads to lower exam scores, suggesting a negative correlation between daily smartphone usage frequency and academic performance (Baert et al., 2020). Additionally, the study by Roig-Merino et al. (2024) revealed a statistically significant negative correlation between smartphone usage time and academic performance. Similarly, Hawi and Samaha (2016) found that smartphone addiction among college students was associated with lower academic achievement, mediated by poor attention regulation and time mismanagement. Kuş (2025), in a recent meta-analysis, further confirmed this pattern by reporting the small but consistent negative effects of problematic smartphone use and gaming on student academic outcomes. Similarly, Mushtaq (2024) found that although students often used smartphones for both social and academic purposes, excessive use—particularly for non-academic activities—leads to significant distractions, resulting in lower cumulative GPA scores. Likewise, another study (Bugti et al., 2023) discovered that smartphone addiction, driven by increased social media use and procrastination, is a key factor in declining academic performance. This indicates that smartphone use may affect the social behavior of university students, thereby impacting their academic performance.

We have discussed four potential reasons behind multitasking or task-switching behaviors associated with smartphone use. First, the visual and auditory notifications on smartphones may capture students' attention during class and/or study periods (Junco & Cotten, 2012), especially notifications from instant messaging platforms. In Taiwan, for instance, notifications from LINE (an instant messaging app) prompt students to frequently check their messages. Second, the desire not to miss out on online happenings and the constant interaction with the outside world (now labeled as "FOMO," or Fear of Missing Out) may cause a lack of focus required for achieving good academic performance (Chen & Yan, 2016). Third, using smartphones during learning-related activities may be the result of addictive behavior and cyberloafing, defined as interference with personal (online) activities during work or study (Vitak et al., 2011). Lastly, due to a lack of motivation to study, students may experience boredom, and smartphone applications offer a quick and tempting form of escape (Hawi & Samaha, 2016).

On the other hand, if used appropriately, smartphones (and their specific functions) can enhance the educational performance of university students. The mobility of smartphones allows students to access internet-based services (similar to those on computers) almost anytime and anywhere (Lepp et al., 2014). Easy access to these functions enables university students to continuously search for information related to their studies. Consequently, smartphones provide a multimedia platform that facilitates learning, which cannot be replaced by simply reading textbooks (Zhang et al., 2014). Dos (2014) indicated that there is no negative correlation because people have already become accustomed to using mobile phones in their daily lives. These varying results may be due to the different purposes of mobile phone use. Lau's (2017) study showed that there is no significant relationship between using mobile phones for academic purposes and student performance, whereas there is a statistically significant negative correlation between using mobile phones for non-academic purposes and academic performance. In a meta-analysis, Sung et al. (2015) demonstrated a moderate positive effect of 0.523 for the application of mobile devices in education. Moreover, social networking sites and messaging apps may facilitate the rapid sharing of relevant information. Faster communication between students, as well as between students and faculty, may foster more effective learning and collaboration. Hashemi et al. (2024) emphasized that students who properly manage their smartphone use and utilize it as an educational resource exhibit better academic performance. Additionally, the study by Cheng et al. (2016) confirmed that using smartphones in the classroom can effectively enhance English language learning among university students, while Traxler (2009) also pointed out that smartphone use can improve learning outcomes. Tessier (2013) found that using smartphones as educational tools can enhance learning effectiveness. Allowing students to use smartphones as learning aids in the classroom can improve academic performance and benefit the educational environment (Tessier, 2013). Moreover, Liebherr et al. (2020) suggested that smartphone use may also improve attention spans, which contrasts with the aforementioned concerns about smartphones diminishing students' attention. Various studies in the literature suggest that academic performance is influenced by issues related to smartphone use and/or smartphone addiction (Ahmed et al., 2020; Amez & Baert, 2020; Baert et al., 2020; Grant et al., 2019; Rathakrishnan et al., 2021; Sunday et al., 2021).

In conclusion, the above arguments suggest that there is sufficient reason to expect a relationship between smartphone use and academic performance among university students. Smartphones support learning environments by providing convenient communication tools and rapid information retrieval. However, the literature thus far has emphasized the potential negative impacts of smartphones, such as distraction, multitasking, and health effects.

2.3. Challenges and opportunities in assessing the impact of smartphone use on academic performance: insights from meta-analysis

From the literature review, it is clear that although the overall trend

suggests a negative correlation between excessive smartphone use and academic performance, the nature of this relationship is multifaceted and depends on factors such as the smartphone usage environment, individual habits, and data collection methods. We believe that through meta-analysis, we can clearly understand whether the impact of smartphone usage frequency on university students' academic performance is positive or negative, and obtain the effect sizes from these studies. Moreover, through subgroup analysis in meta-analysis, we can identify the factors that influence the relationship between smartphone usage frequency and academic performance in the literature, which would be valuable for both academia and policymakers. Finally, meta-analysis focuses on resolving the discrepancies in empirical research findings (and therefore the areas where consensus has not been reached in the literature) and examining how different research designs and methods affect the relationship between smartphone usage frequency and academic performance. This will provide scholars with clearer directions for future productive research.

However, due to the high complexity of smartphones, many studies face difficulties in conducting systematic reviews, which may be attributed to: (a) inconsistent definitions of smartphone usage frequency, (b) varying research designs—some being experimental studies, while others are survey-based studies, (c) discrepancies in the methods used to measure academic performance, as some studies use standardized tests while others use teacher-designed assessments, (d) inconsistent measures of smartphone usage adopted in studies due to the multifunctionality of smartphones, and (e) inconsistent data collection methods, for instance, data obtained through self-reported smartphone usage frequency versus data collected using built-in smartphone tracking applications. These differences can potentially lead to variations in research outcomes. These factors have posed challenges for many researchers attempting systematic reviews in the past.

However, meta-analysis resolves many of the issues encountered in systematic reviews by using standardized scores to estimate effect sizes. Additionally, meta-analysis typically includes both experimental and survey-based studies, acknowledging the diversity of research designs. To address such variability, meta-analysis can conduct subgroup analysis, separating experimental results from survey-based results (Cheung & Slavin, 2016). This allows researchers to compare the effect sizes across different research designs. In other words, the aforementioned issues can be addressed in meta-analysis through subgroup analysis, thereby providing more detailed conclusions (Wang & Fan, 2018). For instance, different measurement tools can be treated as subgroups in the analysis, and this method can reveal whether the relationship between smartphone use and academic performance varies depending on the tools used. Moreover, the inconsistency in data collection methods, which may lead to variations in results, can be addressed through sensitivity analysis during meta-analysis. Sensitivity analysis assesses how the inclusion or exclusion of studies based on data collection methods influences the results (Mathur & VanderWeele, 2020). This approach helps in understanding the robustness of meta-analytic findings and the potential biases introduced by different measurement techniques.

As for the multifunctionality of smartphones, studies often use various measures to assess usage patterns, such as the time spent on social media, gaming, or educational apps. Meta-analysis tackles this issue by coding different types of smartphone activities and examining their distinct impacts on academic performance (Kates et al., 2018). By breaking down smartphone usage into specific categories, researchers can identify which functions have the most significant effects and ensure that the analysis is not biased by an overly broad definition of 'smartphone usage.' Recent studies have echoed this concern by emphasizing the importance of differentiating between smartphone use for academic versus non-academic purposes, and measuring both quantitatively and qualitatively to capture nuanced effects (Kuş, 2025; Sunday et al., 2021).

Although the integration of smartphone usage and academic performance into comprehensive research presents inherent challenges due to variations in definitions, designs, and measurement tools, meta-analysis employs several strategies to overcome these issues. Standardizing categories, conducting subgroup analyses, coding for multifunctionality, and performing sensitivity analyses are key methods that enable researchers to draw more reliable and generalizable conclusions from different studies.

3. Methods

3.1. Inclusion and exclusion criteria

A study is eligible for inclusion in the meta-analysis if it meets the following four criteria:

1. The study's focus is on the impact of smartphone usage frequency on students' academic performance. The article measures some form of smartphone usage frequency, such as: (a) the number/frequency of text messages or phone calls sent/received within a given time frame (the number of times the phone was used in a specific period), (b) the degree of smartphone addiction (measured using a standardized test to assess usage frequency), or (c) the time spent using the phone for various purposes within a specific period. Additionally, the study must address the use of a smartphone, cell phone, or mobile communication device. Studies in which the primary measure is the use of a computer/tablet or the internet are excluded.
2. Students' academic performance is considered the primary dependent variable and is measured through some academic performance structure, such as: (a) GPA, (b) raw exam scores, (c) overall class grades, (d) self-reported academic performance, or (e) performance, difficulty level, or other measures related to any learning tasks or tests constructed by the researchers. Any cognitive tasks that measure learning outcomes, such as hands-on experiments or self-report questionnaires, are also considered as alternative assessments.
3. The study must report sufficient and appropriate data to calculate effect sizes, such as means, standard deviations, t, F, or $\chi 2$ values, or the number of participants in each group. Studies that do not specify sample sizes for each group, lack any inferential statistics, or report inferential statistics but do not provide enough data to calculate effect sizes are excluded.
4. The study must be publicly available online or in library archives.

3.2. Search strategies

The researchers employed a comprehensive search strategy using the following keywords: ("smartphone" OR "mobile phone" OR "cell phone" OR "smart phone") AND ("academic performance" OR "academic achievement" OR "GPA" OR "learning effectiveness" OR "academic outcomes" OR "educational achievement") AND ("college students" OR "university students" OR "undergraduate students").

Boolean operators "AND" and "OR" were used to ensure a broad yet targeted search.

The following databases were searched: *ERIC*, *Education Full Text*, *PsycINFO*, *PubMed*, *ProQuest*, *Google Scholar*, and *Web of Science*.

Additionally, the reference lists of the included studies were reviewed to identify relevant studies not found through electronic searches.

3.3. Coding procedures

A total of **512 articles** were retrieved through the expanded search strategy. These records were imported into a basic coding spreadsheet, and a two-stage screening process was conducted based on pre-established inclusion and exclusion criteria. The **PRISMA flowchart** outlining the study selection process is presented in Fig. 1.

In the **first stage**, we utilized the AI-assisted tool **Scispace** to assist in organizing and filtering the search results. The tool generated lists of

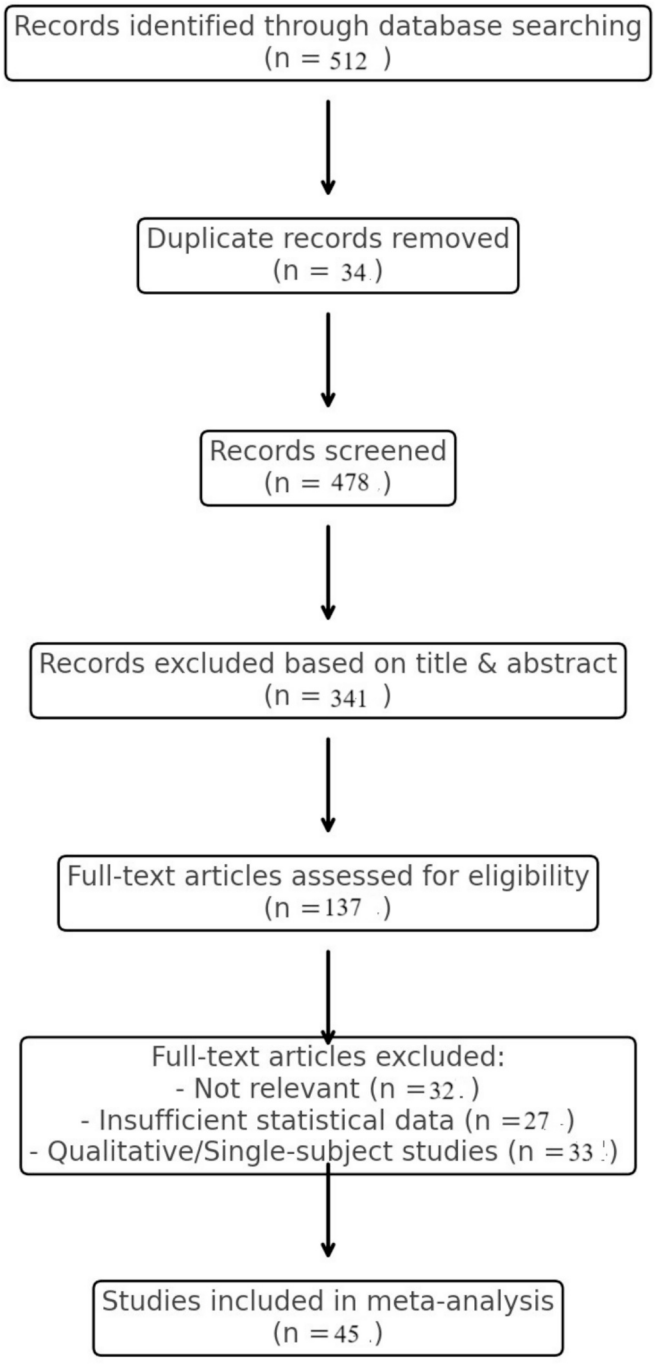


Fig. 1. PRISMA flow diagram of study identification, screening, and inclusion process.

article titles, keywords, and abstracts, which were then manually reviewed by the first author and two trained research assistants (acknowledged in the Acknowledgements section). Each entry was carefully examined to ensure consistency with the inclusion criteria. Following this initial screening, **137 articles** were retained as potentially eligible. In the **second stage**, two independent reviewers assessed the full texts of the 137 studies. Discrepancies in judgment were resolved through discussion. The inter-rater agreement for this screening process was **Po = 0.83**, with a chance-adjusted agreement of **κ = 0.69**, indicating substantial agreement. Ultimately, **45 studies** were included in the final meta-analysis.

For each included study, key characteristics—such as sample size, participant age or grade level, country/region, publication year, and statistical effect size—were extracted and recorded using a structured coding sheet. To ensure the reliability of data extraction, all studies were independently double-coded. The observed agreement for this coding phase was **Po = 0.85**, with a chance-adjusted agreement of **κ = 0.51**, indicating moderate agreement.

3.4. Moderator and mediator analysis

Several characteristics of each sample were coded as potential moderators: (a) source of data (journal or thesis/dissertation), (b) region of the study (Asia, Europe, Americas, and Africa), (c) usage purpose, (d) terminological variations related to smartphone usage frequency, (e) smartphone-related behaviors impacting academic performance, (f) research design (survey or experiment), (g) data collection method (objective measurement or self-reported), and (h) testing tools (standardized tests or researcher-developed tests).

3.5. Data processing and analysis

This study follows the recommendations of Borenstein et al. (2009) and Cooper et al. (2009) for data analysis. Pearson correlation is used as an intuitive method to reflect the impact of smartphone usage frequency on students' academic performance, as it helps in understanding the strength of the relationship between two variables. Therefore, for each outcome measured, researchers calculated the Pearson correlation coefficient (r-value) using the R programming language. However, the correlation coefficients themselves were not used to summarize effect sizes. Instead, the coefficients were transformed into Winer's Z-values, and all analyses were conducted using the transformed values. Finally, the Z-values were converted back into correlation coefficients (Borenstein et al., 2009). In terms of model selection, both fixed-effect and random-effects models were presented in the analysis, but following Borenstein et al.'s (2009) advice, the decision on which model to discuss was based on the results of the Test for Heterogeneity. The R programming language and Python were used to compile effect size data and conduct the analysis.

4. Results

4.1. Overall effect

This meta-analysis synthesized data from **45 studies**, encompassing a total of **84,268 university students** across **18 countries**, including Turkey, Brazil, Japan, India, and Pakistan. The forest plot displaying the **effect sizes (TE)** and **95 % confidence intervals (95 % CI)** for each study is presented in Fig. 2. The individual effect sizes ranged from **−0.26 to + 0.08**.

The overall effect size estimated using the **random-effects model** was **−0.12** (95 % CI: **[−0.13, −0.10]**), indicating a small but statistically significant **negative association** between smartphone usage frequency and academic performance. The corresponding estimate from the **fixed-effects model** was also **−0.11** (95 % CI: **[−0.12, −0.11]**), showing a high degree of consistency between models. This similarity suggests that heterogeneity across studies has minimal influence on the direction and magnitude of the overall effect.

Among the 45 included studies, **38 studies (84.4 %)** reported **negative effect sizes**, most of which had confidence intervals that did not cross zero, suggesting statistically significant negative relationships. In contrast, **3 studies (6.7 %)** reported **positive effect sizes**, but all were small in magnitude and accompanied by wide confidence intervals that included zero, indicating weak and statistically non-significant associations. The remaining **4 studies (8.9 %)** yielded **effect sizes near zero**, with confidence intervals overlapping the null value, suggesting no meaningful association.

The study weights were relatively balanced (ranging from approximately 2.0 % to 2.4 %), indicating no undue influence by any single

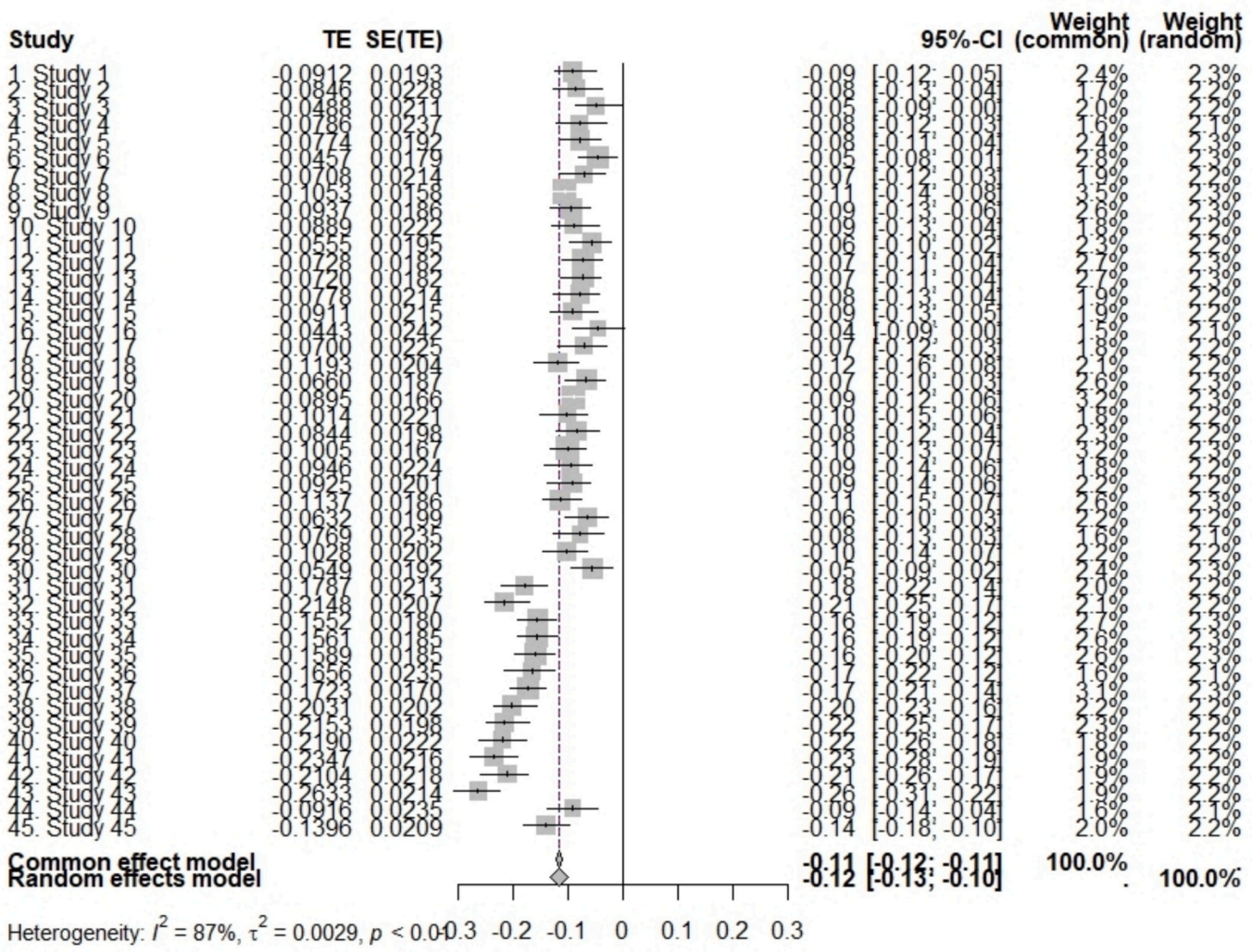


Fig. 2. Forest plot of overall effect meta-analysis.

study. However, the heterogeneity statistic was substantial, with $\boldsymbol{I^2}$ = **87 %**, pointing to considerable variability in effect sizes across studies. This heterogeneity may be attributed to differences in research design, measurement tools, regional or cultural contexts, and participant characteristics. To further explore these sources of variability, a series of **subgroup analyses** was conducted, which are reported in the following section.

4.2. Test for heterogeneity

The heterogeneity of effect sizes was tested under the null hypothesis that all studies share a common true effect size. The results indicate that the overall sample is heterogeneous, as Q(44) = 733.78, $p < .001$. Such results suggest that the null hypothesis (that studies are homogeneous) should be rejected, implying true heterogeneity. This provides evidence that the true effects vary from one study to another. Finally, with $I^2 = 87$ %, it can be inferred that 87 % of the total variability across studies is due to real differences between studies (variance), while 13 % of the variability is due to within-study variance caused by sampling error. Based on these findings, it can be concluded that there are differences in the individual effect sizes that make up the overall result, making it necessary to compare individual variables. This study compares a total of 8 individual variables, as shown in Table 1.

4.3. Test for publication bias and outliers

The asymmetric funnel plot suggests the possibility of publication bias, with a majority of smaller studies clustering to the left of the mean, indicating stronger negative effects of mobile phone use on student academic outcomes. Although visual inspection hints at some asymmetry, a formal Egger's test should be conducted to confirm this bias statistically. Based on the current data, the complete meta-analysis under the random-effects model showed a negative association of −0.12 between mobile phone use and student academic outcomes (95 % CI: −0.13, −0.10). The fixed-effects model produced a similar estimate of −0.11 (95 % CI: −0.12, −0.11), suggesting that the inclusion of more heterogeneous studies does not significantly alter the overall estimate.

Additionally, Rosenthal's fail-safe N calculation could be performed to assess the robustness of the results. For example, if Rosenthal's fail-safe N indicates that a substantial number of unpublished or unidentified studies (e.g., over 4000) would be required to render the observed effect non-significant, it would suggest that the overall conclusions are unlikely to be overturned by missing studies. Given the current dataset of 45 studies, the chance of such a large quantity of unpublished data going unnoticed seems improbable.

Moreover, a separate analysis of the 11 largest studies yielded a consistent result, showing a negative association of −0.12, further confirming the robustness of the meta-analytic findings. The high heterogeneity ($I^2 = 87$ %) indicates variability in study outcomes, likely due

Table 1
Table of group name descriptions for study coding.

Coding	Name	Category (Subcode)
a	Source of data	1. Journal 2. Thesis/dissertation
b	Region of the study	1. Asia 2. Europe 3. Americas 4. Africa
c	Usage purpose	1. Leisure and entertainment 2. Conversations and chatting 3. Sending text messages 4. Contacting family and friends 5. Using social media 6. Playing games and watching videos
d	Terminological variations related to smartphone usage frequency	1. Smartphone addiction/ problematic use 2. Frequency of use 3. Smartphone dependency 4. Total time spent
e	Smartphone-related behaviors impacting academic performance	1. Excessive phone use 2. Excessive texting 3. Multitasking in-class 4. Time spent on phone
f	Research design	1. Survey 2. Experiment
g	Data collection method	1. Objective measurement 2. Self-reported
h	Testing tools	1. Standardized tests 2. Researcher-developed tests

to differences in study design or sample characteristics. Despite this, the overall conclusion remains that mobile phone use is associated with moderately negative academic outcomes for students.

Finally, applying the Trim and Fill method to adjust for potential publication bias might slightly alter the effect size estimate, but it is unlikely to shift the correlation substantially. If asymmetric studies were accounted for, the estimated correlation might adjust to approximately −0.13 to −0.14, similar to the current results, further supporting the consistency of the findings.

4.4. Risk of bias assessment

To address the risk of bias, this meta-analysis included only peer-reviewed quantitative studies that reported sufficient statistical data for calculating effect sizes. Although a formal risk of bias tool such as the Cochrane Risk of Bias or Newcastle-Ottawa Scale was not applied—given the correlational nature of the included studies—several steps were taken to ensure study quality. These include: (1) establishing clear inclusion criteria for quantitative outcomes, (2) assessing publication bias using Egger's regression test and the trim-and-fill method, (3) conducting sensitivity analyses by excluding one study at a time, and (4) performing subgroup analyses based on measurement tools and sample regions. These procedures together help mitigate the potential impact of bias and enhance the credibility of the findings.

4.5. Subgroup analysis

Although the overall effect size provides descriptive value, the I^2 statistic indicates that a large portion of the variance in observed effect sizes is caused by factors beyond sampling error. The observed variability suggests that it is unlikely the same effects will occur under all conditions. Therefore, hypotheses about effect sizes were tested based on the following: (a) source of data, (b) region of the study, (c) usage purpose, (d) terminological variations related to smartphone usage frequency, (e) smartphone-related behaviors impacting academic performance, (f) research design, (g) data collection method, and (h) testing tools. Table 2 summarizes the results of these analyses.

Moderation analysis revealed a statistically significant difference between the weighted average correlation coefficients of unpublished theses/dissertations (r = −0.22) and journal studies (r = −0.11). The between-group homogeneity statistic $Q_b(1) = 30.15$, $p < .05$ (the between-group homogeneity statistic (Q_b) tests for between-group differences, and a Q_b at $p < .05$ indicates statistical significance). The findings suggest that unpublished theses/dissertations produced statistically significant negative effects, which were higher than those of journal studies. Although only four theses/dissertations addressing this topic were collected, the similarity between the fixed effect size and random effect size results suggests that there may be minimal variance between studies. Therefore, it can be claimed that this significant difference aligns with the statistically hypothesized difference. Additionally, it is evident that both journals and theses/dissertations show negative weighted average correlation coefficients, indicating that smartphone usage frequency negatively impacts students' academic performance in both types of studies.

We also examined regional differences in effect sizes. For coding purposes, we categorized studies by continent rather than by country. As shown in Table 2, it is evident that studies from all continents show a negative impact of smartphone usage frequency on students' academic performance. The comparison of between-group differences yielded $Q_b(3) = 21.11$, $p < .05$, indicating significant differences between regions. Africa (r = −0.21), Asia (r = −0.21), Europe (r = −0.16), and the Americas (r = −0.15) show that the weighted average correlation coefficient for Africa is significantly more negative than those of other continents. This finding may be explained by the digital divide, which is discussed further.

We also investigated the purposes of smartphone use. Regarding the purpose of use, the weighted average correlation coefficients for leisure and entertainment (r = −0.02, $p > .05$) and contacting family and friends (r = −0.01, $p > .05$) did not show significant effects. However, conversations and chatting (r = −0.14), sending text messages (r = −0.09), using social media (r = −0.17), and playing games and watching videos (r = −0.20) all had significant weighted average correlation coefficients. The comparison of between-group differences yielded $Q_b(5) = 35.33$, $p < .05$, indicating significant differences across studies. The weighted average correlation coefficients show that using smartphones for playing games and watching videos has the most negative impact on students' academic performance, with a higher negative value than conversations and chatting, sending text messages, and using social media. In contrast, using smartphones for leisure and entertainment or contacting family and friends did not show significant differences.

The study also confirmed that terminological variations related to smartphone usage frequency can moderate the effect size. Based on the literature, no standardized terminology currently exists for smartphone usage frequency. Therefore, this study categorized the frequency terms in the 45 studies into four categories: smartphone addiction/problematic use, frequency of use, smartphone dependency, and total time spent. As shown in Table 2, the between-group difference $Q_b(3) = 12.27$, $p < .05$ was significant, indicating differences between the variables across studies. The 'total time spent' (r = 0.01, $p > .05$) yielded a positive result but was not statistically significant, suggesting that studies using total time spent to measure smartphone usage frequency do not affect students' academic performance. However, the weighted average correlation coefficients for smartphone addiction/problematic use (r = −0.17), smartphone dependency (r = −0.12), and frequency of use (r = −0.13) were all statistically significant and negative, with smartphone addiction/problematic use showing the largest negative value. This indicates that measuring smartphone usage frequency through smartphone addiction/problematic use yields the most significant negative correlation, while frequency of use shows the smallest coefficient. These results suggest that measuring smartphone usage frequency through

Table 2
Summary table of weighted average correlation coefficients for each moderating factor.

Moderator	*k*	Effect size	95 % confidence interval (95 % CI)		Test of heterogeneity		
		r	Lower	Upper	*Q*	*df*	I^2
(Between-levels - source of data)					30.15*	1	
Thesis/dissertation	4	−0.22*	−0.32	−0.17	6.01	3	61.71
Journal	41	−0.11*	−0.12	−0.09	679.87	40	93.33
(Between-levels - region of the study)					21.11*	3	
Africa	5	−0.21*	−0.31	−0.15	0.88	4	0
Asia	16	−0.21*	−0.22	−0.08	441.56*	15	95.99
Europe	5	−0.16*	−0.20	−0.15	52.58*	4	93.16
Americas	19	−0.15*	−0.18	−0.09	231.71*	18	90.09
(Between-levels - usage purpose)					35.33*	5	
Leisure and entertainment	4	−0.02	−0.07	0.01	3.99*	3	70.01
Contacting family and friends	2	−0.01	−0.06	0.05	3.33*	1	65.55
Conversations and chatting	3	−0.14*	−0.16	−0.06	10.88*	2	66.00
Sending text messages	16	−0.09*	−0.11	−0.04	303.36*	15	93.78
Using social media	18	−0.17*	−0.19	−0.13	381.24*	17	96.94
Playing games and watching videos	2	−0.20*	−0.25	−0.11	0.32	1	0
(Between-levels - terminological variations related to smartphone usage frequency)					12.27*	3	
Smartphone addiction/problematic use	14	−0.17*	−0.19	−0.10	556.34*	13	96.17
Frequency of use	18	−0.12*	−0.15	−0.09	144.08*	17	89.01
Smartphone dependency	12	−0.13*	−0.16	−0.08	13.39*	11	81.62
Total time spent	1	0.01	−0.07	0.11	0	0	0
(Between-levels - smartphone-related behaviors impacting academic performance)					25.88*	3	
Multitasking in-class	9	−0.22*	−0.26	−0.15	20.21*	8	60.11
Excessive texting	11	−0.08*	−0.12	−0.05	201.33*	10	96.33
Time spent on phone	8	−0.15*	−0.17	−0.12	333.40*	7	95.27
Excessive phone use	16	−0.09*	−0.10	−0.07	109.92*	15	90.09

* $p < .05$.

smartphone addiction/problematic use, smartphone dependency, and frequency of use tends to yield results indicating a negative impact on academic performance.

Another important moderating variable is smartphone-related behaviors that impact academic performance. The literature suggests that most studies tend to view smartphone use as having a negative impact on students' academic performance. In this study, the overall effect size was also negative, but given the $I^2 = 87$ %, there are likely several potential moderating variables. We are particularly interested in which smartphone-related behaviors students engage in during class that truly affect their academic performance. As shown in Table 2, multitasking in-class, excessive texting, time spent on the phone, and excessive phone use all produced negative results. The between-group comparison yielded $Q_b(3) = 25.88$, $p < .05$, indicating significant differences in the weighted correlation coefficients between the variables. The weighted average correlation coefficients show that multitasking in-class (r = −0.22) has a significantly higher negative impact than excessive texting (r = −0.08), time spent on the phone (r = −0.15), and excessive phone use (r = −0.09). This suggests that students' multitasking during class has a more negative impact on academic performance than excessive texting, time spent on the phone, or excessive phone use.

Table 3 summarizes the analysis of weighted average correlation coefficients for Research Objective 3, which aimed to examine whether research design, data collection method, and testing tools functioned as moderating variables. Regarding research design, Table 3 shows that only survey-based studies (r = −0.10) had a statistically significant difference, while experimental studies (r = −0.01) did not. The between-group comparison yielded $Q_b(1) = 2.61$, $p > .05$, which was not statistically significant, indicating no difference between survey-based and experimental studies.

Regarding data collection methods, both objective measurement (r = −0.16) and self-reported data (r = −0.09) produced negative results, with objective measurement showing a larger negative value. The between-group comparison yielded $Q_b(1) = 4.77$, $p < .05$, indicating statistical significance. These results suggest that using objective measurement methods to collect data results in a larger negative effect on academic performance compared to self-reported methods when measuring smartphone usage frequency.

Finally, in terms of testing tools, both researcher-developed tests (r = −0.13) and standardized tests (r = −0.10) produced negative results, with researcher-developed tests showing a higher negative value. The between-group comparison yielded $Q_b(1) = 9.13$, $p < .05$, indicating statistical significance. These results suggest that researcher-developed tests are more likely to yield negative effects of smartphone usage frequency on students' academic performance, with a larger negative value than standardized tests.

Table 3
Summary table of weighted average correlation coefficients for moderating factors of experimental design and methodology.

Moderator	*k*	Effect size	95 % confidence interval (95 % CI)		Test of heterogeneity		
		r	Lower	Upper	*Q*	*df*	I^2
(Between-levels - research design)					2.61	1	
Survey-study	42	−0.10*	−0.12	−0.09	633.66*	41	93.40
Experiment-study	3	−0.01	−0.07	0.02	80.34*	2	89.93
(Between-levels - data collection method)					4.77*	1	
Objective measurement	32	−0.16*	−0.18	−0.16	633.71*	31	92.10
Self-reported	13	−0.09*	−0.11	−0.01	80.35*	12	93.34
(Between-levels - testing tools)					9.13*	1	
Researcher-developed test	26	−0.13*	−0.15	−0.10	510.40*	25	92.24
Standardized tests	19	−0.10*	−0.13	−0.05	199.07*	18	92.89

* $p < .05$.

5. Summary of results and discussion

This meta-analysis aims to investigate the extent to which smartphone usage frequency impacts university students' academic performance, and if so, to what degree this impact is influenced by factors such as (a) source of data, (b) region of the study, (c) usage purpose, (d) terminological variations related to smartphone usage frequency, and (e) smartphone-related behaviors impacting academic performance. Additionally, it seeks to determine how different research designs and methodologies—including research design, data collection methods, and testing tools—affect the relationship between smartphone usage frequency and university students' academic performance.

The findings of this study indicate that overall, smartphone usage frequency has a small negative impact on university students' academic performance ($r = -0.12$), consistent with previous literature (Hawi & Samaha, 2016; Kates et al., 2018; Lepp et al., 2015; Sunday et al., 2021). While prior research has primarily focused on smartphone addiction and general usage, this study concentrates more specifically on usage frequency. Smartphone addiction implies overuse, while general usage often reflects widespread survey data due to the increased prevalence of smartphones. Smartphone usage frequency, positioned between these two, helps fill a gap in the existing literature. Given the negative correlation found in this study, it can be inferred that for every unit increase in smartphone usage frequency, there is a 0.14 unit decrease in academic performance. However, this result should not be oversimplified, as the relationship between smartphone usage frequency and academic performance may be non-linear (Straker et al., 2022). For instance, Gorjón and Osés (2023) pointed out that the relationship between ICT use and academic performance follows an inverted U-shape, where overuse leads to poorer performance. This suggests that moderate use can be beneficial, but overuse is harmful, further complicating the relationship.

Additionally, results from the Test for publication bias and outliers indicate that these findings are not significantly biased due to the systematic exclusion of studies. However, it should be noted that the observed effects may suggest an association rather than a causal relationship. For example, individuals who tend to overuse mobile devices may not have been likely to achieve higher academic performance in the first place.

Moreover, based on the results of the subgroup analysis, factors such as source of data, region of the study, usage purpose, terminological variations related to smartphone usage frequency, and smartphone-related behaviors impacting academic performance were examined in the moderation analysis. Statistically significant differences were found between groups, with negative impacts across the board. The detailed discussion of each factor follows.

(a) **Source of data**: In terms of data sources, both unpublished theses/dissertations and journal articles yielded negative results. However, it is worth noting that the weighted average correlation coefficient for unpublished theses was higher than for journal articles. This could be due to the relatively less rigorous sampling, testing, or experimental designs in unpublished theses compared to journal articles.

(b) **Region of the study**: Studies conducted in Asia, the Americas, Africa, and Europe all consistently show that smartphone usage frequency has a negative impact on university students' academic performance, with the highest weighted average correlation coefficient found in studies from Africa. Researchers speculate that this result may be related to the digital divide. Numerous studies have confirmed that disparities in digital access affect students' learning outcomes. For instance, students' inability to access stable internet, educational apps, or appropriate digital tools can lead to poorer academic performance (Liu, 2023; Zhou & Deng, 2024). The lack of access to digital resources may exacerbate the negative impact of smartphone usage, as students may rely more heavily on their devices for educational purposes, leading to distraction and lower academic performance (Liu, 2023; Zhou & Deng, 2024). For example, Darko-Adjei's (2019) study investigated the smartphone usage of distance learning students at an African university. While smartphones were found to play an important role in academic activities, issues such as unstable internet connections and phone malfunctions during study sessions disrupted learning and negatively impacted academic outcomes. Our subgroup analyses revealed that the negative association between smartphone use and academic performance was more pronounced in studies conducted in Asian regions compared to those in North America or Europe. In addition to infrastructure-related factors such as the digital divide, cultural and educational expectations may also contribute to these regional differences. In many East Asian societies, academic success is heavily emphasized due to highly competitive educational systems and strong familial expectations rooted in Confucian cultural values such as filial piety, obedience, and respect for authority (Chao, 1994; Ho, 1981; Shek, 2002). Parents are more likely to adopt performance-oriented and controlling parenting styles to ensure their children's academic achievement (Ng et al., 2014; Pong et al., 2010). As a result, distractions caused by smartphone use may be perceived as more detrimental and more directly associated with academic underperformance in these contexts. In contrast, Western educational cultures often emphasize self-regulated learning, student autonomy, and holistic development. The integration of digital tools such as smartphones into teaching and learning is more common and may be perceived as beneficial rather than disruptive. These cultural, pedagogical, and policy-related differences likely contribute to the observed regional variation and underscore the importance of culturally contextualized interpretation in cross-cultural meta-analytic research.

(c) **Usage purpose**: Among the variables, only *Leisure and entertainment* and *Contacting family and friends* showed no significant effects in their weighted average correlation coefficients. However, the coefficients for *Conversations and chatting*, *Sending text messages*, *Using social media*, and *Playing games and watching videos* were significant. These findings are consistent with previous studies. For example, Dietz and Henrich's (2014) research pointed out that texting during class negatively impacts academic performance. Similarly, Rozgonjuk et al. (2018) found that students who are absorbed in social media experience negative effects on their learning performance. This result suggests that the impact of smartphone usage frequency on academic performance depends on the purpose of use. For instance, some individuals may benefit from spending hours each day talking to friends, while others who use social media apps for only a few minutes during study or sleep time may experience negative effects (Sohn et al., 2021).

(d) **Terminological variations related to smartphone usage frequency**: Studies showed that using *Total Time Spent* to measure the impact of smartphone usage frequency on students' academic performance yielded positive but non-significant results. This suggests that using *Total Time Spent* as a measurement method cannot effectively capture the impact of smartphone usage on academic performance. This result is particularly interesting, as it reaffirms the previous point: "The impact of smartphone usage frequency on academic performance depends on the purpose of use." If we only consider the total time students spend using smartphones, we cannot determine what activities they are engaging in on their devices, and thus cannot establish a clear correlation with academic performance. For example, some students may spend a significant amount of time using their smartphones to complete assignments or study, while others may spend the same amount of time browsing social media or engaging in

entertainment. If we use total time as the only measurement criterion, the results for these two groups of students would be entirely different, making it impossible to accurately reflect the true impact of smartphone usage frequency on academic performance. Additionally, when using the terms *Smartphone Addiction/Problematic Use*, *Frequency of use*, and *Smartphone dependency* to measure the impact of smartphone usage frequency on academic performance, the results were all negative, with *Smartphone Addiction/Problematic Use* having the highest weighted correlation coefficient. This indicates that using stronger terms like "Addiction" and "Problematic Use" tends to yield more pronounced negative results, as these terms inherently carry negative connotations. Amez and Baert (2020) support this view, arguing that studies focusing on addiction-related behaviors are more likely to find substantial negative correlations with academic success, likely due to the embedded negative meanings of these terms.

(e) **Smartphone-related behaviors impacting academic performance**: In this section, we found that *Multitasking in class*, *Excessive texting*, *Time spent on phone*, and *Excessive phone use* all negatively impacted university students' academic performance, with *Multitasking in class* having the highest weighted correlation coefficient. This result is consistent with previous studies, such as those by Junco and Cotten (2012) and Jacobsen and Forste (2011), which indicated that multitasking with smartphones during class, although convenient, tends to distract students and negatively impact their academic performance. Additionally, Dos (2014) found that excessive texting negatively affects learning. The digital divide may also contribute to these outcomes; for instance, when students multitask during class, issues such as unstable internet access, lack of educational apps, and inadequate digital tools can cause frustration, further lowering their learning outcomes (Liu, 2023; Zhou & Deng, 2024). Furthermore, self-discipline and self-regulation may play a crucial role. Smartphone use in class can be easily influenced by notifications from social media or instant messaging, reducing focus. For example, in Muñoz et al.'s (2019) study, 45 % of students reported using their smartphones for chatting during class, and 29 % accessed social media during class. Bonneville and Riddell (2023) explored self-regulation strategies to mitigate the risks of smartphone use during class. Their study found that while smartphones are often portrayed negatively in the media, some students consciously adopt strategies like turning off notifications or putting their phones away to avoid distractions. Despite these efforts, the lure of social media notifications remains a challenge for many students (Bonneville & Riddell, 2023). While many studies emphasize distraction and reduced attention, some findings suggest that under certain conditions, moderate smartphone use may actually improve attentional engagement, as shown in Liebherr et al. (2020).

This study also analyzed whether research design, data collection method, and testing tools served as moderating variables. The results indicated no significant differences between survey-based studies and experimental studies. Regarding data collection methods, both objective measurements and self-reported data yielded negative results, with objective measurements showing higher values. This may be due to the fact that self-reported smartphone usage is easily influenced by social desirability and recall bias, as individuals may not accurately remember or may underreport their actual screen time. For example, Geyer et al. (2022) pointed out that tools such as open-source smartphone applications provide detailed and customizable usage data, offering more accurate insights into human-computer interaction than self-reports. Self-reported measures often fail to accurately reflect actual smartphone usage. A study found a weak correlation between self-reported and objectively measured social media use, indicating potential inaccuracies in self-reported data (Mahalingham et al., 2023). Hartanto et al. (2023) also suggested that compared to self-reported data, objective measurements, particularly those tracking smartphone-checking behavior, may provide more accurate insights into the negative effects of smartphone use on cognitive function. Große Deters and Schoedel (2024) compared self-reported and objective measures regarding the impact of excessive smartphone use on health and found that while objective measurements yielded negative results, the self-reported results were not significant. This suggests that self-reports tend to be more biased than objective measurements, especially when assessing smartphone usage frequency.

Finally, in terms of testing tools, the results showed that both researcher-developed tests and standardized tests yielded negative outcomes, with researcher-developed tests showing higher values. This indicates that standardized tests may have higher validity and reliability compared to researcher-developed tests.

5.1. Practical implications for educators and policymakers

While the findings of this study highlight the importance of digital literacy and self-regulation in mitigating the negative impact of smartphone use on academic performance, actionable strategies are needed to translate these insights into practice. Educators could consider implementing structured smartphone-use guidelines in the classroom, such as designated phone-free periods, or promoting mobile-free learning zones to minimize distractions during instruction (Flanigan & Babchuk, 2023).

Schools and universities may also benefit from incorporating digital literacy and self-regulation training into their curricula, enabling students to better manage notifications, screen time, and multitasking behaviors. At the policy level, educational authorities could support these efforts by developing national or institutional frameworks for responsible smartphone use. These targeted strategies would help maximize the benefits of mobile technology while minimizing its academic costs.

6. Conclusions and implications

This meta-analysis provides important insights into how smartphone usage frequency affects university students' academic performance. The overall findings reveal a small but significant negative impact ($r = -0.12$), reaffirming previous research that has highlighted the detrimental effects of increased smartphone usage on academic performance. Importantly, this study fills a gap in the literature by specifically focusing on smartphone usage frequency, which is distinct from broader concepts such as smartphone addiction or general smartphone use. Thus, it allows for a more nuanced understanding of the incremental impact of frequent smartphone use on students' academic outcomes.

6.1. Implications for research and practice

1. **Theoretical Implications**: This study emphasizes the importance of distinguishing between different measures of smartphone use—such as addiction, dependency, and usage frequency—when analyzing academic outcomes, thereby enriching the knowledge base. The results indicate that usage frequency negatively impacts academic performance, but the extent of this impact depends on the characteristics of the usage. For instance, terms like "Smartphone Addiction" or "Problematic Use" tend to produce stronger negative correlations compared to "Total Time Spent." This distinction highlights the need for consistent definitions and terminology in future research to improve the comparability of findings.

   Additionally, previous studies (Straker et al., 2022; Gorjón & Osés, 2023) have suggested the existence of a potential non-linear relationship between smartphone use and academic performance, which merits further investigation. Moderate smartphone use may enhance learning, while excessive use is harmful, following an inverted U-shaped pattern. Future research should continue to explore the threshold at which smartphone use becomes detrimental to academic

performance, potentially focusing on different academic and cultural contexts.

2. **Practical Implications**: Educators and policymakers should take note of the significant variation in how smartphone use affects students, depending on the "purpose of use." The negative impacts of activities such as gaming, social media, and excessive texting are particularly evident. By contrast, using smartphones for leisure, entertainment, or contacting family and friends does not seem to produce significant adverse effects. This underscores the need for targeted interventions that help students manage their smartphone usage more effectively, particularly in contexts where distractions from social media and other non-educational activities are prevalent.

   Government and university institutions should consider promoting information literacy programs that guide university students on how to plan and use smartphones effectively in academic settings. Encouraging students to adopt practical strategies such as self-control/self-regulation may help mitigate the disruptive effects of multitasking with smartphones during class. These measures might include turning off notifications, using time management apps, or limiting smartphone use for educational purposes during study periods. Many apps now allow users to set when they are activated or deactivated, which could be particularly helpful for students who have not yet adopted self-control/self-regulation practices—this could also be an area for future research.
3. **Methodological Contributions**: The findings regarding research design and data collection methods also carry significant implications. Objective measures of smartphone use (such as app-based tracking) provide more reliable data than self-reported usage, which is prone to recall bias and social desirability effects. Therefore, future studies examining the effects of smartphone use on academic performance should prioritize objective data collection methods to improve the accuracy of their findings.

   Furthermore, standardized tests seem to provide more reliable and valid insights when compared to researcher-developed tools, which tend to produce more pronounced negative effects. This suggests that future research should favor standardized assessments of academic performance to ensure the consistency and replicability of results.

6.2. Limitations and areas for future research

While this meta-analysis provides valuable insights, several limitations must be acknowledged. The high heterogeneity (I^2 = 87 %) indicates substantial variability among the included studies, which may be due to differences in methodologies, cultural contexts, or definitions of smartphone use. Future meta-analyses should consider addressing this heterogeneity through more granular subgroup analyses, particularly focusing on cultural, educational, and technological factors that may moderate the relationship between smartphone use and academic performance.

Moreover, although this study found a negative correlation between smartphone usage frequency and academic performance, the results cannot establish causality. The observed relationship may be influenced by confounding variables, such as study design, smartphone usage purpose, and regional disparities, as identified in our subgroup analysis. Longitudinal studies are needed to explore whether frequent smartphone use leads to lower academic performance or whether students with lower academic performance are more prone to excessive smartphone use. Additionally, future research should adopt stronger causal inference methodologies, such as **randomized controlled trials (RCTs), propensity score matching (PSM), instrumental variable approaches, and difference-in-differences (DiD) analysis**, to mitigate unmeasured confounding effects and strengthen causal interpretations. Furthermore, researchers should examine whether the negative impact of smartphone use is moderated by specific activities (e.g., academic vs. non-academic use), and whether these effects vary across different educational settings. Such methodological advancements will contribute to a more rigorous understanding of the causal mechanisms underlying the relationship between smartphone use and academic performance.

The high level of heterogeneity (I^2 = 87 %) observed in this meta-analysis reflects not only statistical variation but also conceptual and methodological inconsistencies across studies. Smartphone use, as a multidimensional and socially embedded behavior, has been described using a wide range of terms—such as *addiction*, *dependency*, *problematic use*, and *frequent use*—often without clear definitions or consistent operational criteria. This inconsistency complicates comparisons and may inflate heterogeneity.

In our analysis, we adopted "usage frequency" as the primary criterion to identify studies for inclusion, due to its relative availability and comparability. However, distinguishing between behavioral frequency and psychological dependence remains essential. Furthermore, the development of standardized, psychometrically validated tools is needed to reliably measure levels of smartphone involvement and their potential educational impacts.

On the outcome side, academic performance has traditionally been assessed through the GPA or standardized tests. Yet these tools may not fully capture the subtle cognitive, motivational, or behavioral changes that smartphone use may produce. In addition to limitations associated with self-reported smartphone usage, many of the included studies also relied on **self-reported academic performance metrics**, such as GPA. These measures are subject to potential biases, including **social desirability effects and recall inaccuracies**, which may either inflate or attenuate the observed associations depending on participants' perceptions and willingness to report honestly. This further contributes to measurement-related heterogeneity.

Future studies should prioritize the use of **objective academic performance indicators**, such as institutional grade records or standardized test scores, to enhance validity. We also recommend that researchers (1) clearly define the operational criteria for smartphone-related constructs, (2) distinguish between frequency of use and dependence-related symptoms, and (3) consider adopting **multidimensional frameworks** that encompass cognitive, emotional, and behavioral components. Such efforts would improve measurement consistency, enhance meta-analytic comparability, and advance theoretical development in this growing field.

Finally, the study underscores the need for a more standardized approach to measuring smartphone use in academic contexts. Researchers should work toward developing comprehensive frameworks that account for both usage frequency and purpose, ensuring a more holistic understanding of how digital behaviors influence learning.

To further address the limitations identified in this study, we recommend that future research adopt **longitudinal designs**—such as repeated-measures or multi-wave panel studies—to clarify causal directions and developmental changes in smartphone use and academic performance over time. Such designs could help disentangle whether frequent smartphone use leads to poorer academic outcomes or whether lower-achieving students are more prone to excessive use.

In addition, researchers should work toward developing **standardized, psychometrically validated instruments** that differentiate between smartphone usage frequency, problematic behaviors, and psychological dependency. These tools should be culturally adaptable and sensitive to context-specific learning environments to enhance cross-study comparability.

Finally, future research would benefit from incorporating **objective measurement strategies** for both smartphone use (e.g., digital trace data, smartphone tracking apps) and academic performance (e.g., institutional records, standardized assessments), which would help reduce biases associated with self-reporting and increase the reliability and validity of findings.

In conclusion, this study highlights that frequent smartphone use is associated with lower academic performance among university students, with the specific effects depending on the purpose of the use,

behavioral patterns, and research methods. As smartphones continue to play a central role in students' lives, educators and policymakers must prioritize interventions that help students use smartphones effectively and appropriately. By fostering information literacy and encouraging self-control/self-regulation in smartphone use, university institutions can help students navigate the challenges posed by technology and optimize their academic potential.

CRediT authorship contribution statement

Chia-Yen Hsieh: Writing – original draft, Validation, Methodology, Funding acquisition, Data curation, Software, Resources, Formal analysis, Writing – review & editing, Visualization, Project administration, Investigation, Conceptualization, Supervision.

Informed consent

All participants in this study have completed an informed consent form.

Ethical approval

The study was approved by the National Cheng Kung University Governance Framework for Human Research Ethics. The approval number is NCKU HREC-E-110-327-2. And it has been confirmed that the experiment has obtained informed consent from all patients/participants.

Ethics statement

This study was approved by the institutional review board.

Statement regarding research involving human participants and/or animals

Not applicable

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used [Open AI] to [enhance the English writing] in order to [improve readability, as English is not our first language]. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of this publication.

Funding

The work done for this study was financially supported by the Ministry of Science and Technology of Taiwan under project No. MOST 110-2511-H-259-009.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to express their gratitude to the school participants in the study.

Thank you, Dr. Wang, for collaborating with me in screening articles for the meta-analysis.

Appendix A

Appendix Table 1
Summary of included studies.

Author & year	Sample size	Region	Study design	Smartphone use measurement	Academic performance measure
Jacobsen and Forste (2011)	1026	Americas	Survey-study	Smartphone dependency	GPA
Junco and Cotten (2012)	1839	Americas	Survey-study	Smartphone dependency	GPA
Junco (2012)	1774	Americas	Survey-study	Frequency of use	GPA
Beland and Murphy (2016)	130	Europe	Survey-study	Smartphone addiction	Researcher-developed test
Lee (2015)	304	Americas	Survey-study	Smartphone addiction	GPA
Jackson et al. (2011)	491	Americas	Survey-study	Frequency of use	GPA
Lepp et al. (2014)	536	Americas	Survey-study	Frequency of use	Actual GPA
Lepp et al. (2015)	536	Americas	Survey-study	Frequency of use	Actual GPA
Chen and Ji (2015)	506	Asia	Survey-study	Frequency of use	Actual GPA
Olufadi (2015a, 2015b)	1000	Africa	Survey-study	Frequency of use	Researcher-developed test
Alosaimi et al. (2016)	2367	Asia	Survey-study	Frequency of use	Standardized tests
Baert et al. (2020)	1786	Europe	Survey-study	Smartphone addiction	Researcher-developed test
Chen and Peng (2008)	26,307	Asia	Survey-study	Smartphone addiction	GPA
Christensen et al. (2016)	653	Americas	Survey-study	Frequency of use	Researcher-developed test
Felisoni and Godoi (2018)	43	Americas	Experiment-study	Frequency of use	Standardized tests
Janković et al. (2016)	485	Europe	Survey-study	Frequency of use	Researcher-developed test
Kim et al. (2019)	84	Asia	Experiment-study	Frequency of use	Researcher-developed test
Li et al. (2015)	516	Americas	Survey-study	Frequency of use	Researcher-developed test
Nayak (2018)	429	Asia	Survey-study	Total time spent	Standardized tests
Asante and Hiadzi (2018)	150	Africa	Survey-study	Smartphone addiction	GPA
Lin and Chiang (2017)	438	Asia	Survey-study	Smartphone dependency	Researcher-developed test
Rashid and Asghar (2016)	761	Asia	Survey-study	Smartphone addiction	Researcher-developed test
Rosen et al. (2018)	216	Americas	Survey-study	Frequency of use	Researcher-developed test
Samaha and Hawi (2016)	293	Asia	Survey-study	Smartphone addiction	GPA
Sert et al. (2019)	743	Asia	Survey-study	Smartphone addiction	Researcher-developed test
Uzun and Kilis (2019)	631	Asia	Survey-study	Smartphone dependency	Researcher-developed test
Wentworth and Middleton (2014)	483	Americas	Survey-study	Frequency of use	Researcher-developed test
Winskel et al. (2019)	389	Asia	Survey-study	Frequency of use	Researcher-developed test

(*continued on next page*)

Appendix Table 1 (*continued*)

Author & year	Sample size	Region	Study design	Smartphone use measurement	Academic performance measure
Ibrahim et al. (2018)	610	Asia	Survey-study	Smartphone addiction	GPA
Olufadi (2015a, 2015b)	286	Africa	Survey-study	Smartphone dependency	Researcher-developed test
Witecki and Nonnecke (2015)	972	Americas	Survey-study	Smartphone dependency	Researcher-developed test
Dos (2014)	250	Asia	Survey-study	Smartphone dependency	Researcher-developed test
Grant et al. (2019)	31,425	Americas	Survey-study	Frequency of use	GPA
Matar Boumosleh and Jaalouk (2017)	688	Asia	Survey-study	Smartphone dependency	Researcher-developed test
Katz and Lambert (2016)	104	Americas	Experiment-study	Smartphone addiction	Researcher-developed test
Lin et al. (2021)	2900	Asia	Survey-study	Frequency of use	Standardized tests
Nayak (2018)	429	Asia	Survey-study	Smartphone addiction	Researcher-developed test
Khan et al. (2019)	360	Asia	Survey-study	Smartphone dependency	GPA
Hawi and Samaha (2016)	293	Asia	Survey-study	Smartphone addiction	GPA
Ifeanyi and Chukwuere (2018)	375	Africa	Survey-study	Smartphone addiction	Researcher-developed test
Mokoena (2012)	375	Africa	Survey-study	Smartphone dependency	Researcher-developed test
Kumar et al. (2021)	370	Asia	Survey-study	Smartphone dependency	Researcher-developed test
Sharma et al. (2022)	436	Asia	Survey-study	Frequency of use	Researcher-developed test
Edeh (2024)	221	Africa	Survey-study	Smartphone dependency	Researcher-developed test
Amez and Baert (2020)	1673	Europe	Survey-study	Smartphone addiction	Researcher-developed test

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

References

Ahmed, R. R., Salman, F., Malik, S. A., Streimikiene, D., Soomro, R. H., & Pahi, M. H. (2020). Smartphone use and academic performance of university students: A mediation and moderation analysis. *Sustainability, 12*(1), 439. https://doi.org/10.3390/su12010439

Aker, S., Şahin, M. K., Sezgin, S., & Oğuz, G. (2017). Psychosocial factors affecting smartphone addiction in university students. *Journal of Addictions Nursing, 28*(4), 215–219. https://doi.org/10.1097/JAN.0000000000000197

Al-Khlaiwi, T. M., Habib, S. S., Meo, S. A., Alqhtani, M. S., & Ogailan, A. A. (2020). The adult population. Association of smart mobile phone usage with cognitive function impairment in Saudi adult population. *Pakistan Journal of Medical Sciences, 36*(7). https://doi.org/10.12669/pjms.36.7.2826

Alosaimi, F. D., Alyahya, H., Alshahwan, H., Al Mahyijari, N., & Shaik, S. A. (2016). Smartphone addiction among university students in Riyadh, Saudi Arabia. *Saudi Medical Journal, 37*(6), 675. https://doi.org/10.15537/smj.2016.6.14430

Amez, S., & Baert, S. (2020). Smartphone use and academic performance: A literature review. *International Journal of Educational Research, 103*, Article 101618. https://doi.org/10.1016/j.ijer.2020.101618

Anshari, M., Almunawar, M. N., Shahrill, M., Wicaksono, D. K., & Huda, M. (2017). Smartphones usage in the classrooms: Learning aid or interference? *Education and Information Technologies, 22*, 3063–3079. https://doi.org/10.1007/s10639-017-9572-7

Asante, R. K., & Hiadzi, R. A. (2018). In-lecture smartphone use and academic performance: A reflection on the sustainable development goal number four. *Ghana Social Science Journal, 15*(2), 161–178.

Aydin, M. K., & Kuş, M. (2023). Nomophobia and smartphone addiction amidst COVID-19 home confinement: The parallel mediating role of digital gaming and social media tools usage across secondary school students. *Frontiers in Psychology, 14*, Article 1175555. https://doi.org/10.3389/fpsyg.2023.1175555

Baert, S., Vujić, S., Amez, S., Claeskens, M., Daman, T., Maeckelberghe, A., … De Marez, L. (2020). Smartphone use and academic performance: Correlation or causal relationship? *Kyklos, 73*(1), 22–46. https://doi.org/10.1111/kykl.12214

Beland, L. P., & Murphy, R. (2016). Ill communication: Technology, distraction & student performance. *Labour Economics, 41*, 61–76. https://doi.org/10.1016/j.labeco.2016.04.004

Bianchi, A., & Phillips, J. G. (2005). Psychological predictors of problem mobile phone use. *CyberPsychology and Behavior, 8*(1), 39–51. https://doi.org/10.1089/cpb.2005.8.39

Billieux, J., Maurage, P., Lopez-Fernandez, O., Kuss, D. J., & Griffiths, M. D. (2015). Can disordered mobile phone use be considered a behavioral addiction? An update on current evidence and a comprehensive model for future research. *Current Addiction Reports, 2*(2), 156–162. https://doi.org/10.1007/s40429-015-0054-y

Bonneville, L., & Riddell, D. (2023). Smartphones in the university classroom: Less problematic than we tend to think? *Information, Communication & Society, 26*(5), 1008–1022. https://doi.org/10.1080/1369118X.2023.2166358

Borenstein, M., Hedges, L. V., Higgins, J. P. T., & Rothstein, H. R. (2009). *Introduction to meta-analysis.* Wiley.

Bragazzi, N. L., Re, T. S., & Zerbetto, R. (2019). The relationship between nomophobia and maladaptive coping styles in a sample of Italian young adults: Insights and implications from a cross-sectional study. *JMIR Mental Health, 6*(4), Article e13154. https://doi.org/10.2196/13154

Bugti, S. M., Bugti, F., & Sarhandi, P. S. A. (2023). Determinants of smartphone addiction and its influence on students' academic performance. *Pakistan Journal of Distance Learning, 9*(2), 34–45. https://doi.org/10.30971/pjdol.v9i2.1894

Bulut, A., & Sengul, H. (2024). The moderating role of gender in the relationship between nomophobia and social interaction anxiety in university students. *International Journal of Human Computer Interaction, 40*(14), 3583–3596. https://doi.org/10.1080/10447318.2023.2191079

Busch, P. A., & McCarthy, S. (2021). Antecedents and consequences of problematic smartphone use: A systematic literature review of an emerging research area. *Computers in Human Behavior, 114*, Article 106414. https://doi.org/10.1016/j.chb.2020.106414

Chao, R. K. (1994). Beyond parental control and authoritarian parenting style: Understanding Chinese parenting through the cultural notion of training. *Child Development, 65*, 1111–1119. https://doi.org/10.1111/j.1467-8624.1994.tb00806.x

Chen, J., Liang, Y., Mai, C., Zhong, X., & Qu, C. (2016). General deficit in inhibitory control of excessive smartphone users: Evidence from an event-related potential study. *Frontiers in Psychology, 7*, 511. https://doi.org/10.3389/fpsyg.2016.00511

Chen, Q., & Yan, Z. (2016). Does multitasking with mobile phones affect learning? A review. *Computers in Human Behavior, 54*, 34–42. https://doi.org/10.1016/j.chb.2015.07.047

Chen, R. S., & Ji, C. H. (2015). Investigating the relationship between thinking style and personal electronic device use and its implications for academic performance. *Computers in Human Behavior, 52*, 177–183. https://doi.org/10.1016/j.chb.2015.05.042

Chen, Y. F., & Peng, S. S. (2008). University students' Internet use and its relationships with academic performance, interpersonal relationships, psychosocial adjustment, and self-evaluation. *Cyberpsychology & Behavior, 11*(4), 467–469. https://doi.org/10.1089/cpb.2007.0128

Cheng, Y. M., Kuo, S. H., Lou, S. J., & Shih, R. C. (2016). The development and implementation of u-msg for college students' English learning. *International Journal of Distance Education Technologies, 14*(2), 17–29. https://doi.org/10.4018/IJDET.2016040102

Cheung, A., & Slavin, R. E. (2016). How methodological features affect effect sizes in education. *Educational Researcher, 45*(5), 275–284. https://doi.org/10.3102/0013189X16656615

Christensen, M. A., Bettencourt, L., Kaye, L., Moturu, S. T., Nguyen, K. T., Olgin, J. E., … Marcus, G. M. (2016). Direct measurements of smartphone screen-time: Relationships with demographics and sleep. *PloS One, 11*(11), Article e0165331. https://doi.org/10.1371/journal.pone.0165331

Cooper, H. M., Hedges, L. V., & Valentine, J. (Eds.). (2009). *The handbook of research synthesis and meta-analysis* (2nd ed.). Russell Sage Foundation. https://doi.org/10.7758/9781610448864.

Darko-Adjei, N. (2019). The use and effect of smartphones in students' learning activities: Evidence from the University of Ghana, Legon. *Journal of Educational Technology, 24*(2), 143–155.

Demirci, K., Akgönül, M., & Akpinar, A. (2015). Relationship of smartphone use severity with sleep quality, depression, and anxiety in university students. *Journal of Behavioral Addictions, 4*(2), 85–92. https://doi.org/10.1556/2006.4.2015.010

De-Sola Gutiérrez, J., Rodríguez de Fonseca, F., & Rubio, G. (2016). Cell-phone addiction: A review. *Frontiers in Psychiatry, 7*, 175. https://doi.org/10.3389/fpsyt.2016.00175

Dietz, S., & Henrich, C. (2014). Texting as a distraction to learning in college students. *Computers in Human Behavior, 36*, 163–167. https://doi.org/10.1016/j.chb.2014.03.045

Dixit, S., Shukla, H., Bhagwat, A. K., Bindal, A., Goyal, A., Zaidi, A. K., & Shrivastava, A. (2010). A study to evaluate mobile phone dependence among students of a medical college and associated hospital of central India. *Indian Journal of Community Medicine, 35*(2), 339–341. https://doi.org/10.4103/0970-0218.66878

Dos, B. (2014). The relationship between mobile phone use, metacognitive awareness and academic achievement. *European Journal of Educational Research, 3*(4), 192–200. https://doi.org/10.12973/eu-jer.3.4.192

Edeh, F. C. (2024). Smartphone usage and academic performance of Management Science students in an emerging economy. *International Journal of Applied Business and International Management, 9*(1), 62–74. https://doi.org/10.32535/ijabim.v9i3.3593

Felisoni, D. D., & Godoi, A. S. (2018). Cell phone usage and academic performance: An experiment. *Computers & Education, 117*, 175–187. https://doi.org/10.1016/j.compedu.2017.10.006

Flanigan, A. E., & Babchuk, W. A. (2023). Managing student digital distraction in the college classroom: A self-determination theory perspective. *Educational Psychology Review, 35*(60). https://doi.org/10.1007/s10648-023-09780-y

Geng, Y., Gu, J., Wang, J., & Zhang, R. (2021). Smartphone addiction and depression, anxiety: The role of bedtime procrastination and self-control. *Journal of Affective Disorders, 293*, 415–421. https://doi.org/10.1016/j.jad.2021.06.062

Geyer, K., Ellis, D. A., Shaw, H., & Davidson, B. I. (2022). Open-source smartphone app and tools for measuring, quantifying, and visualizing technology use. *Behavior Research Methods, 54*(1), 1–12. https://doi.org/10.3758/s13428-021-01585-7

Gorjón, L., & Osés, A. (2023). The negative impact of information and communication technologies overuse on student performance: Evidence from OECD countries. *Journal of Educational Computing Research, 61*(4), 723–765. https://doi.org/10.1177/07356331221133408

Grant, J. E., Lust, K., & Chamberlain, S. R. (2019). Problematic smartphone use associated with greater alcohol consumption, mental health issues, poorer academic performance, and impulsivity. *Journal of Behavioral Addictions, 8*(2), 335–342. https://doi.org/10.1556/2006.8.2019.32

Große Deters, F., & Schoedel, R. (2024). Keep on scrolling? Using intensive longitudinal smartphone sensing data to assess how everyday smartphone usage behaviors are related to well-being. *Computers in Human Behavior, 150*, Article 107977. https://doi.org/10.1016/j.chb.2023.107977

Guo, Z., Lu, X., Li, Y., & Li, Y. (2011). A framework of students' reasons for using CMC media in learning contexts: A structural approach. *Journal of the Association for Information Science and Technology, 62*(11), 2182–2200. https://doi.org/10.1002/asi.21631

Gutiérrez-Puertas, L., Márquez-Hernández, V. V., São-Romão-Preto, L., Granados-Gámez, G., Gutiérrez-Puertas, V., & Aguilera-Manrique, G. (2019). Comparative study of nomophobia among Spanish and Portuguese nursing students. *Nurse Education in Practice, 34*, 79–84. https://doi.org/10.1016/j.nepr.2018.11.010

Harris, B., Regan, T., Schueler, J., & Fields, S. A. (2020). Problematic mobile phone and smartphone use scales: A systematic review. *Frontiers in Psychology, 11*, 672. https://doi.org/10.3389/fpsyg.2020.00672

Hartanto, A., Lee, K. Y., Chua, Y. J., Quek, F. Y., & Majeed, N. M. (2023). Smartphone use and daily cognitive failures: A critical examination using a daily diary approach with objective smartphone measures. *British Journal of Psychology, 114*(1), 70–85. https://doi.org/10.1111/bjop.12597

Hashemi, A., Noori, A. Q., Orfan, S. N., Akramy, S. A., & Rameli, M. R. M. (2024). Undergraduate students' perception of smartphone addiction and its impact on their academic performance: A case study. *Cogent Education.*. https://doi.org/10.1080/2331186x.2024.2340845

Haug, S., Castro, R. P., Kwon, M., Filler, A., Kowatsch, T., & Schaub, M. P. (2015). Smartphone use and smartphone addiction among young people in Switzerland. *Journal of Behavioral Addictions, 4*(4), 299–307. https://doi.org/10.1556/2006.4.2015.037

Hawi, N. S., & Samaha, M. (2016). To excel or not to excel: Strong evidence on the adverse effect of smartphone addiction on academic performance. *Computers & Education, 98*, 81–89. https://doi.org/10.1016/j.compedu.2016.03.007

Ho, D. Y. (1981). Traditional patterns of socialization in Chinese society. *Acta Psychological Taiwanica, 23*, 81–95. https://doi.org/10.2307/352376

Ho, R. C., Zhang, M. W., Tsang, T. Y., Toh, A. H., Pan, F., Lu, Y., … Mak, K. K. (2014). The association between internet addiction and psychiatric co-morbidity: A meta-analysis. *BMC Psychiatry, 14*, 1–10. https://doi.org/10.1186/1471-244X-14-183

Ibrahim, N. K., Baharoon, B. S., Banjar, W. F., Jar, A. A., Ashor, R. M., Aman, A. A., & Al-Ahmadi, J. R. (2018). Mobile phone addiction and its relationship to sleep quality and academic achievement of medical students at King Abdulaziz University, Jeddah, Saudi Arabia. *Journal of Research in Health Sciences, 18*(3), Article e00420.

Ifeanyi, I. P., & Chukwuere, J. E. (2018). The impact of using smartphones on the academic performance of undergraduate students. *Knowledge Management & E-Learning, 10*(3), 290–308.

Jackson, L. A., Von Eye, A., Fitzgerald, H. E., Witt, E. A., & Zhao, Y. (2011). Internet use, videogame playing and cell phone use as predictors of children's body mass index (BMI), body weight, academic performance, and social and overall self-esteem. *Computers in Human Behavior, 27*(1), 599–604. https://doi.org/10.1016/j.chb.2010.10.019

Jacobsen, W., & Forste, R. (2011). The wired generation: Academic and social outcomes of electronic media use among university students. *Cyberpsychology, Behavior, and Social Networking, 14*(5), 275–280. https://doi.org/10.1089/cyber.2010.0135

Janković, B., Nikolić, M., Vukonjanski, J., & Terek, E. (2016). The impact of Facebook and smart phone usage on the leisure activities and college adjustment of students in Serbia. *Computers in Human Behavior, 55*, 354–363. https://doi.org/10.1016/j.chb.2015.09.022

Junco, R. (2012). In-class multitasking and academic performance. *Computers in Human Behavior, 28*(6), 2236–2243. https://doi.org/10.1016/j.chb.2012.06.031

Junco, R., & Cotten, S. R. (2012). No A 4 U: The relationship between multitasking and academic performance. *Computers & Education, 59*(2), 505–514. https://doi.org/10.1016/j.compedu.2011.12.023

Kang, S., & Jung, J. (2014). Mobile communication for human needs: A comparison of smartphone use between the US and Korea. *Computers in Human Behavior, 35*, 376–387. https://doi.org/10.1016/j.chb.2014.03.024

Karikoski, J., & Soikkeli, T. (2013). Contextual usage patterns in smartphone communication services. *Personal and Ubiquitous Computing, 17*(3), 491–502. https://doi.org/10.1007/s00779-011-0503-0

Kates, A. W., Wu, H., & Coryn, C. L. (2018). The effects of mobile phone use on academic performance: A meta-analysis. *Computers & Education, 127*, 107–112. https://doi.org/10.1016/j.compedu.2018.08.012

Katz, L., & Lambert, W. (2016). A happy and engaged class without cell phones? It's easier than you think. *Teaching of Psychology, 43*(4), 340–345. https://doi.org/10.1177/0098628316662767

Khan, A. A., Khalid, A., & Iqbal, R. (2019). Revealing the relationship between smartphone addiction and academic performance of students: Evidences from higher educational Institutes of Pakistan. *Pakistan Administrative Review, 3*(2), 74–83.

Khayer, E., Zarei, R., Damghanian, M., Bőthe, B., & Farnam, F. (2024). Compulsive sexual behaviour in Iranian married women: Prevalence, sociodemographic, sexual, and psychological predictors across-country. *Journal of Behavioral Addictions*. https://doi.org/10.1556/2006.2024.00009

Kim, I., Kim, R., Kim, H., Kim, D., Han, K., Lee, P. H., … Lee, U. (2019). Understanding smartphone usage in college classrooms: A long-term measurement study. *Computers & Education, 141*, Article 103611. https://doi.org/10.1016/j.compedu.2019.103611

Kim, S., & Byrne, S. (2011). Conceptualizing personal web usage in work contexts: A preliminary framework. *Computers in Human Behavior, 27*, 2271–2283. https://doi.org/10.1016/j.chb.2011.07.006

Kumar, P. R. S., Aruna, K., Kumar, A., & Venkatalakshmi, P. (2021). A smartphone use and its impact on academic performance of medical students: A cross sectional study. *International Journal of Advances in Medicine, 8*(10), 1582–1587. https://doi.org/10.18203/2349-3933.ijam20213714

Kuş, M. (2025). A meta-analysis of the impact of technology related factors on students' academic performance. *Frontiers in Psychology, 16*, Article 1524645. https://doi.org/10.3389/fpsyg.2025.1524645

Kuss, D. J., & Griffiths, M. D. (2017). Social networking sites and addiction: Ten lessons learned. *International Journal of Environmental Research and Public Health, 14*(3), 311. https://doi.org/10.3390/ijerph14030311

Kuss, D. J., Kanjo, E., Crook-Rumsey, M., Kibowski, F., Wang, G. Y., & Sumich, A. (2018). Problematic mobile phone use and addiction across generations: The roles of psychopathological symptoms and smartphone use. *Journal of Technology in Behavioral Science, 3*, 141–149. https://doi.org/10.1007/s41347-017-0041-3

Kwon, M., Kim, D. J., Cho, H., & Yang, S. (2013). The smartphone addiction scale: Development and validation of a short version for adolescents. *PLoS One, 8*(12), Article e83558. https://doi.org/10.1371/journal.pone.0083558

Kwon, M., Lee, J. Y., Won, W. Y., Park, J. W., Min, J. A., Hahn, C., … Kim, D. J. (2013). Development and validation of a smartphone addiction scale (SAS). *PLoS One, 8*(2), Article e56936. https://doi.org/10.1371/journal.pone.0056936

Lau, W. W. (2017). Effects of social media usage and social media multitasking on the academic performance of university students. *Computers in Human Behavior, 68*, 286–291. https://doi.org/10.1016/j.chb.2016.11.043

Lee, E. B. (2015). Too much information: Heavy smartphone and Facebook utilization by African American young adults. *Journal of Black Studies, 46*(1), 44–61. https://doi.org/10.1177/0021934714557035

Lee, H.-K., Kim, J.-H., Fava, M., Mischoulon, D., Park, J.-H., Shim, E.-J., Lee, E.-H., Lee, J.-H., & Jeon, H.-J. (2017). Development and validation study of the smartphone Overuse Screening questionnaire. *Psychiatry Research, 257*, 352–357. https://doi.org/10.1016/j.psychres.2017.07.074

Lee, J., Cho, B., Kim, Y., & Noh, J. (2015). Smartphone addiction in university students and its implication for learning. In G. Chen, V. K. Kinshuk, R. Huang, & S. C. Kong (Eds.), *Emerging issues in smart learning* (pp. 297–305). Springer. https://doi.org/10.1007/978-3-662-44188-6_40.

Lepp, A., Barkley, J. E., & Karpinski, A. C. (2014). The relationship between cell phone use, academic performance, anxiety, and satisfaction with life in college students. *Computers in Human Behavior, 31*, 343–350. https://doi.org/10.1016/j.chb.2013.10.049

Lepp, A., Barkley, J. E., & Karpinski, A. C. (2015). The relationship between cell phone use and academic performance in a sample of US college students. *SAGE Open, 5*(1), Article 2158244015573169. https://doi.org/10.1177/2158244015573169

Li, J., Lepp, A., & Barkley, J. E. (2015). Locus of control and cell phone use: Implications for sleep quality, academic performance, and subjective well-being. *Computers in Human Behavior, 52*, 450–457. https://doi.org/10.1016/j.chb.2015.06.021

Liebherr, M., Schubert, P., Antons, S., Montag, C., & Brand, M. (2020). Smartphones and attention, curse or blessing? A review on the effects of smartphone usage on attention, inhibition, and working memory. *Computers in Human Behavior Reports, 1*, Article 100005. https://doi.org/10.1016/j.chbr.2020.100005

Lin, T. T., & Chiang, Y. H. (2017). Investigating predictors of smartphone dependency symptoms and effects on academic performance, improper phone use and perceived sociability. *International Journal of Mobile Communications, 15*(6), 655–676. https://doi.org/10.1504/IJMC.2017.086881

Lin, Y., Liu, Y., Fan, W., Tuunainen, V. K., & Deng, S. (2021). Revisiting the relationship between smartphone use and academic performance: A large-scale study. *Computers in Human Behavior, 122*, Article 106835. https://doi.org/10.1016/j.chb.2021.106835

Liu, C. (2023). The unique role of smartphone addiction and related factors among university students: A model based on cross-sectional and cross-lagged network analyses. *BMC Psychiatry, 23*(1), 883. https://doi.org/10.1186/s12888-023-05384-6

Lukoff, K., Yu, C., Kientz, J., & Hiniker, A. (2018). What makes smartphone use meaningful or meaningless? *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 2*(1), 1–26. https://doi.org/10.1145/3191754

Mahalingham, T., McEvoy, P. M., & Clarke, P. J. (2023). Assessing the validity of self-report social media use: Evidence of no relationship with objective smartphone use. *Computers in Human Behavior, 140*, Article 107567. https://doi.org/10.1016/j.chb.2022.107567

Matar Boumosleh, J., & Jaalouk, D. (2017). Depression, anxiety, and smartphone addiction in university students-A cross sectional study. *PloS one, 12*(8), Article e0182239. https://doi.org/10.1371/journal.pone.0182239

Mathur, M. B., & VanderWeele, T. J. (2020). Sensitivity analysis for unmeasured confounding in meta-analyses. *Journal of the American Statistical Association, 115*(529), 163–172. https://doi.org/10.1080/01621459.2018.1529598

Mokoena, S. (2012). *Smartphones and regular cellular phones: assessing their impact on students' education at the University of Zululand* [Master's thesis, University of Zululand].

Muñoz, C. F., Almaraz, I. A., & García, F. G. (2019). The effects of spontaneous use of smartphone on the attention and emotions of millennials students in a classroom. *International Linguistics Research, 2*(2), 42–51. https://doi.org/10.30560/ilr.v2n2p42

Mushtaq, I. (2024). The impact of smartphones on the academic performance of university students. *ARETL Journal, 1*(1), 15–27. https://doi.org/10.33422/aretl.v1i1.188

Nayak, J. K. (2018). Relationship among smartphone usage, addiction, academic performance and the moderating role of gender: A study of higher education students in India. *Computers & Education, 123*, 164–173. https://doi.org/10.1016/j.compedu.2018.05.007

Ng, F. F. Y., Pomerantz, E. M., & Deng, C. (2014). Why are Chinese mothers more controlling than American mothers? "My child is my report card". *Child Development, 85*(1), 355–369. https://doi.org/10.1111/cdev.12102

Nguyen, T. V., Nguyen, Q. A. N., Nguyen, N. P., & Nguyen, U. B. (2024). Smartphone use, nomophobia, and academic achievement in Vietnamese high school students. *Computers in Human Behavior Reports, 14*, Article 100418. https://doi.org/10.1016/j.chbr.2024.100418

Olson, J. A., Sandra, D. A., Chmoulevitch, D., Raz, A., & Veissière, S. P. (2023). A nudge-based intervention to reduce problematic smartphone use: Randomised controlled trial. *International Journal of Mental Health and Addiction, 21*(6), 3842–3864. https://doi.org/10.31234/osf.io/tjynk

Olson, J. A., Sandra, D. A., Colucci, É. S., Al Bikaii, A., Chmoulevitch, D., Nahas, J., … Veissière, S. P. (2022). Smartphone addiction is increasing across the world: A meta-analysis of 24 countries. *Computers in Human Behavior, 129*, Article 107138. https://doi.org/10.1016/j.chb.2021.107138

Olufadi, Y. (2015a). A configurational approach to the investigation of the multiple paths to success of students through mobile phone use behaviors. *Computers & Education, 86*, 84–104. https://doi.org/10.1016/j.compedu.2015.03.005

Olufadi, Y. (2015b). Gravitating towards mobile phone (GoToMP) during lecture periods by students: Why are they using it? and how can it be measured? *Computers & Education, 87*, 423–436. https://doi.org/10.1016/j.compedu.2015.08.013

Panova, T., & Carbonell, X. (2018). Is smartphone addiction really an addiction? *Journal of Behavioral Addictions, 7*(2), 252–259. https://doi.org/10.1556/2006.7.2018.49

Park, J.-H., Kim, J.-H., Kim, J.-G., Kim, K.-G., Kim, N.-K., Choi, I.-S., Lee, S.-J., & Yim, J.-G. (2015). The effects of heavy smartphone use on the cervical angle, pain threshold of neck muscles and depression. *Advanced Science and Technology Letters, 91*, 12–17. https://doi.org/10.14257/ASTL.2015.91.03

Park, N., & Lee, H. (2012). Social implications of smartphone use: Korean college students' smartphone use and psychological well-being. *Cyberpsychology, Behavior, and Social Networking, 15*(9), 491–497. https://doi.org/10.1089/cyber.2011.0580

Pong, S. L., Johnston, J., & Chen, V. (2010). Authoritarian parenting and Asian adolescent school performance: Insights from the US and Taiwan. *International Journal of Behavioral Development, 34*(1), 62–72. https://doi.org/10.1177/0165025409345073

Rajeshkumar, C., & Rajakumari, K. (2024, July). Statistical analysis on the effect of Internet-based education in student communities during COVID-19. In *2024 2nd international conference on sustainable computing and smart systems (ICSCSS)* (pp. 315–320). IEEE. https://doi.org/10.1109/icscss60660.2024.10625467.

Rashid, T., & Asghar, H. M. (2016). Technology use, self-directed learning, student engagement and academic performance: Examining the interrelations. *Computers in Human Behavior, 63*, 604–612. https://doi.org/10.1016/j.chb.2016.05.084

Rathakrishnan, B., Bikar Singh, S. S., Kamaluddin, M. R., Yahaya, A., Mohd Nasir, M. A., Ibrahim, F., & Ab Rahman, Z. (2021). Smartphone addiction and sleep quality on academic performance of university students: An exploratory research. *International Journal of Environmental Research and Public Health, 18*(16), 8291. https://doi.org/10.3390/ijerph18168291

Roig-Merino, B., Sigalat-Signes, E., Miret-Pastor, L., & Suárez-Guerrero, C. (2024). University students' perceptions of how mobile phones influence their academic performance: A case study in Valencia, Spain. *Journal of Infrastructure, Policy and Development, 8*(8), 57–79. https://doi.org/10.24294/jipd.v8i8.5779

Rosen, L. D., Carrier, L. M., Pedroza, J. A., Elias, S., O'Brien, K. M., Lozano, J., … Ruiz, A. (2018). The role of executive functioning and technological anxiety (FOMO) in college course performance as mediated by technology usage and multitasking habits. *Psicología Educativa, 24*(1), 14–21. https://doi.org/10.5093/psed2018a3

Rozgonjuk, D., Saal, K., & That, K. (2018). Problematic smartphone use, deep and surface approaches to learning, and social media use in lectures. *International Journal of Environmental Research and Public Health, 15*(1), 92. https://doi.org/10.3390/ijerph15010092

Samaha, M., & Hawi, N. S. (2016). Relationships among smartphone addiction, stress, academic performance, and satisfaction with life. *Computers in Human Behavior, 57*, 321–325. https://doi.org/10.1016/j.chb.2015.12.045

Sanchez-Fernandez, M., & Borda-Mas, M. (2023). Problematic smartphone use and specific problematic Internet uses among university students and associated predictive factors: A systematic review. *Education and Information Technologies, 28*(6), 7111–7204. https://doi.org/10.1007/s10639-022-11437-2

Sarker, I. H., Kayes, A. S. M., & Watters, P. (2019). Effectiveness analysis of machine learning classification models for predicting personalized context-aware smartphone usage. *Journal of Big Data, 6*, 57. https://doi.org/10.1186/s40537-019-0219-y

Satchell, L. P., Fido, D., Harper, C. A., Shaw, H., Davidson, B., Ellis, D. A., … Pavetich, M. (2021). Development of an Offline-Friend Addiction Questionnaire (O-FAQ): Are most people really social addicts? *Behavior Research Methods, 53*, 1097–1106. https://doi.org/10.3758/s13428-020-01462-9

Sert, H., Taskin Yilmaz, F., Karakoc Kumsar, A., & Aygin, D. (2019). Effect of technology addiction on academic success and fatigue among Turkish university students. *Fatigue: Biomedicine, Health & Behavior, 7*(1), 41–51. https://doi.org/10.1080/21641846.2019.1585598

Sha, L., Looi, C. K., Chen, W., & Zhang, B. H. (2012). Understanding mobile learning from the perspective of self-regulated learning. *Journal of Computer Assisted Learning, 28*(4), 366–378. https://doi.org/10.1111/j.1365-2729.2011.00461.x

Sharma, R., Vaidya, V., Rajan, R., & Eldhose, A. T. (2022). Smartphone dependency and its impact on academics among medical and nursing students: A cross-sectional study. *Journal of Integrative Nursing, 4*(1), 30–35. https://doi.org/10.4103/jin.jin_44_21

Shek, D. T. (2002). Parenting characteristics and parent-adolescent conflict: A longitudinal study in the Chinese culture. *Journal of Family Issues, 23*(2), 189–208. https://doi.org/10.1177/0192513X02023002002

Sohn, D. (2014). Coping with information in social media: The effects of network structure and knowledge on perception of information value. *Computers in Human Behavior, 32*, 145–151. https://doi.org/10.1016/j.chb.2013.12.006

Sohn, S. Y., Krasnoff, L., Rees, P., Kalk, N. J., & Carter, B. (2021). The association between smartphone addiction and sleep: A UK cross-sectional study of young adults. *Frontiers in Psychiatry, 12*, Article 629407. https://doi.org/10.3389/fpsyt.2021.629407

Sohn, S. Y., Rees, P., Wildridge, B., Kalk, N. J., & Carter, B. (2019). Prevalence of problematic smartphone usage and associated mental health outcomes amongst children and young people: A systematic review, meta-analysis and GRADE of the evidence. *BMC Psychiatry, 19*, 1–10. https://doi.org/10.1186/s12888-019-2350-x

Steel, C. (2012). Fitting learning into life: Language students' perspectives on benefits of using mobile apps. Retrieved from https://www.ascilite.org/conferences/Wellington12/2012/images/custom/steel,_caroline_-_fitting_learning.pdf.

Straker, L. M., Harris, C. J., Joosten, J., & Howie, E. K. (2022). Academic implications of screen use and sedentary behaviour in a school with a 1-to-1 device policy. *Technology, Pedagogy and Education, 31*(4), 525–539. https://doi.org/10.1080/1475939x.2022.2050288

Sunday, O. J., Adesope, O. O., & Maarhuis, P. L. (2021). The effects of smartphone addiction on learning: A meta-analysis. *Computers in Human Behavior Reports, 4*, Article 100114. https://doi.org/10.1016/j.chbr.2021.100114

Sung, Y. T., Chang, K. E., & Yang, J. M. (2015). How effective are mobile devices for language learning? A meta-analysis. *Educational Research Review, 16*, 68–84. https://doi.org/10.1016/j.edurev.2015.09.001

Suresh, A., Sudhan, S., Mohan, P., & Ramalingam, A. T. (2021). Impact of smartphone addiction on neck pain and disability in university students. *Journal of Clinical and Diagnostic Research, 15*(6), 1–3. https://doi.org/10.7860/JCDR/2021/49339.15029

Tessier, J. (2013). Student impressions of academic cellphone use in the classroom. *Journal of College Science Teaching, 43*(1), 25–29. https://doi.org/10.2505/4/jcst13_043_01_25

Thornton, P., & Houser, C. (2005). Using mobile phones in English education in Japan. *Journal of Computer Assisted Learning, 21*(3), 217–228. https://doi.org/10.1111/j.1365-2729.2005.00129.x

Traxler, J. (2009). Learning in a mobile age. *International Journal of Mobile and Blended Learning, 1*(1), 1–12. https://doi.org/10.4018/jmbl.2009010101

Uzun, A. M., & Kilis, S. (2019). Use in relation to multitasking, self-regulation and academic performance. *Computers in Human Behavior, 90*, 196–203. https://doi.org/10.1016/j.chb.2018.08.045

Van Deursen, A. J. A. M., Bolle, C. L., Hegner, S. M., & Kommers, P. A. M. (2015). Modeling habitual and addictive smartphone behavior: The role of smartphone usage types, emotional intelligence, social stress, self-regulation, age, and gender. *Computers in Human Behavior, 45*, 411–420. https://doi.org/10.1016/j.chb.2014.12.039

Vitak, J., Crouse, J., & LaRose, R. (2011). Personal Internet use at work: Understanding cyberslacking. *Computers in Human Behavior, 27*(5), 1751–1759. https://doi.org/10.1016/j.chb.2011.03.002

Wang, M., Shen, R., Novak, D., & Pan, X. (2009). The impact of mobile learning on students' learning behaviours and performance: Report from a large blended classroom. *British Journal of Educational Technology, 40*(4), 673–695. https://doi.org/10.1111/j.1467-8535.2008.00846.x

Wang, Y., & Fan, X. (2018). Testing group differences using structural equation modeling. *Structural Equation Modeling: A Multidisciplinary Journal, 25*(1), 1–18. https://doi.org/10.1080/10705511.2017.1387859

Wentworth, D. K., & Middleton, J. H. (2014). Technology use and academic performance. *Computers & Education, 78*, 306–311. https://doi.org/10.1016/j.compedu.2014.06.012

Wilmer, H. H., Sherman, L. E., & Chein, J. M. (2017). Smartphones and cognition: A review of research exploring the links between mobile technology habits and

cognitive functioning. *Frontiers in Psychology, 8*, 605. https://doi.org/10.3389/fpsyg.2017.00605
Winskel, H., Kim, T. H., Kardash, L., & Belic, I. (2019). Smartphone use and study behavior: A Korean and Australian comparison. *Heliyon, 5*(7), Article e02158. https://doi.org/10.1016/j.heliyon.2019.e02158
Witecki, G., & Nonnecke, B. (2015). Engagement in digital lecture halls: A study of student course engagement and mobile device use during lecture. *Journal of Information Technology Education. Research, 14*, 73. Retrieved from http://www.jite.org/documents/Vol14/JITEv14ResearchP073-090Witecki0720.pdf.
Yang, Z., Asbury, K., & Griffiths, M. D. (2021). A cancer in the minds of youth? A qualitative study of problematic smartphone use among undergraduate students. *International Journal of Mental Health and Addiction, 19*(4), 934–946. https://doi.org/10.1007/s11469-019-00204-z
Yi, Y. J., You, S., & Bae, B. J. (2016). The influence of smartphones on academic performance: The development of the technology-to-performance chain model. *Library Hi Tech, 34*(3), 480–499. https://doi.org/10.1108/LHT-04-2016-0038
Yildirim, C., & Correia, A. P. (2015). Exploring the dimensions of nomophobia: Development and validation of a self-reported questionnaire. *Computers in Human Behavior, 49*, 130–137. https://doi.org/10.1016/j.chb.2015.02.059
Zhang, G., Yang, X., Tu, X., Ding, N., & Lau, J. T. F. (2020). Prospective relationships between mobile phone dependence and mental health status among Chinese undergraduate students with college adjustment as a mediator. *Journal of Affective Disorders, 260*, 498–505. https://doi.org/10.1016/j.jad.2019.09.047
Zhang, J., & Zeng, Y. (2024). Effect of college students' smartphone addiction on academic achievement: The mediating role of academic anxiety and moderating role of sense of academic control. *Psychology Research and Behavior Management*, 933–944. https://doi.org/10.2147/PRBM.S442924
Zhang, M. W., Ho, C. S., & Ho, R. (2014). Methodology of development and students' perceptions of a psychiatry educational smartphone application. *Technology and Health Care, 22*(6), 847–855. https://doi.org/10.3233/THC-140861
Zhou, Y., & Deng, L. (2024). Breaking free from the "digital rabbit hole": A configurational analysis of in-class smartphone distraction among university students. *The Internet and Higher Education, 62*, Article 100949. https://doi.org/10.1016/j.iheduc.2024.100949

Chia-Yen Hsieh is an Associate Professor and Chair of the Department of Early Childhood Education at National Pingtung University, Taiwan. His interdisciplinary expertise encompasses special education, gifted education, early childhood education, STEAM education, educational neuroscience, and applied statistics. His recent research focuses on the intersection of neuroscience and early childhood education, with an emphasis on inclusive pedagogies and equity in educational opportunities.Dr. Hsieh actively engages with the United Nations Sustainable Development Goals (SDGs), particularly SDG 4 (Quality Education), SDG 10 (Reduced Inequalities), SDG 3 (Good Health and Well-being), and SDG 11 (Sustainable Cities and Communities). His work places a special focus on the development and well-being of children in rural and underserved communities, including Indigenous populations. Through both research and teaching, he advocates for the educational rights of marginalized groups and addresses the digital divide via practice-based learning initiatives.